

PRAHARI

Integrated Video Management & Analytics Platform

પ્રહરી · प्रहरी — “the one who keeps watch”

Technical Proposal — High-Level Design (HLD)

Mandatory submission document 2 of the Sentinel submission requirements

Submitted by	Team PRAHARI
Team lead	Agraj Singh
Members	[Members: ...]
Category	[Category: ...]
Document version	1.0 (first submission)
Date	28 September 2026
Source code	Repository submitted with this document; every “built” claim names the file that implements it

Declared model: Hybrid. Model 1 (registry and GIS) as the mandatory foundation; Model 3 (federation middleware) as the integration spine; Model 2 (direct connection) for each camera or recorder; and Model 4’s analytics — ANPR, watchlist correlation, cross-camera tracking — *without* Model 4’s central video transport, storage and GPU cost.

Core principle: move observations, not video. Video stays in the department that owns it; only small, structured events cross the State.

Contents

1	Executive summary	3
2	Problem understanding and constraints H8	4
3	Choice of model and justification model justification	5
4	Overall architecture H1 · D1	7
4.2	Statewide tiers: edge, regional, central	7
4.3	Component view	8
4.4	End-to-end data flow: plate read → alert → trace	9
4.5	Reference implementation and production target	9
5	Integrating heterogeneous cameras, NVRs and VMS H2 · D2	9
6	Ingesting and managing streams from dispersed sites H3 · S3	12
7	Watchlist integration and real-time correlation H4 · D2	13
8	AI video analytics H5 · D3 · S2	15
9	Alert generation and notification workflow H6	18
10	Cybersecurity, privacy and evidence integrity H7 · D4 · S6	19
11	Deployment architecture D5 · S1	22
12	Infrastructure sizing and cost-benefit D6 · D7 · S1–S4 · S7	23
13	Scalability, availability and disaster recovery H7 · D9 · S5 · S6	29
14	Department-wise information requirements and prerequisites H8 · D8	30
15	Statewide rollout plan and future roadmap D10 · R1	32
16	Working system and test evidence H7 performance	33
A	Appendix A — API endpoint reference	35
B	Appendix B — Data contracts and database schema	36
C	Appendix C — Requirement traceability matrix all requirements	37
D	Appendix D — Glossary and references	39

Grey codes are the official requirements each section answers: H1–H8 = HLD content items (Submission Requirements, document 2); D1–D10 = solution design dimensions (Step 3); S1–S7 = scalability items (Step 6); R1 = rollout plan. Appendix C is the full traceability matrix.

Document control

Title	PRAHARI — Technical Proposal: High-Level Design (HLD)
Version / status	1.0 — submitted for the Sentinel shortlisting round
Answers	Submission requirement “2. Technical Proposal — High-Level Design (HLD)” (eight items), the ten solution-design dimensions of Step 3 and the seven scalability items of Step 6 of the official Problem Statement. Appendix C maps every official requirement to the section that answers it, with build status.
Source of truth	Where this document and the submitted code could differ, the code is authoritative . Every capability marked BUILT is implemented in the repository and exercised by its test suite or by the practice-mode run described in Section 16.
Companion artefacts	Solution presentation (PPT/PDF); own-feed and government-feed demonstration videos; plates output report (CSV); live API documentation at /docs of the running platform.

Reading conventions

This is a design for a statewide system, written by a team that has also built a working reference implementation. To keep the two apart, every capability carries one status label and every number one provenance label.

BUILT	Implemented in the submitted code and tested; the implementing file is named.	measured	From running the submitted code; run, machine and date given.
DESIGNED	Specified here (interfaces, behaviour, parameters), not yet implemented; target of the pilot.	derived	Calculated from measurements and/or stated assumptions; formula or source given.
ROADMAP	Planned for a later rollout phase; intent only.	assumed	An input assumption, listed so it can be challenged and replaced by a surveyed value.

Spelling follows Indian English. Times are Indian Standard Time (UTC+05:30) unless stated. “Camera” includes a channel of an NVR/DVR or VMS. All monetary figures come from the team’s sizing model (Section 12): ₹, excluding GST, accurate to about ±30%.

1 Executive summary

The problem. Twenty-six departments run their own CCTV — analog and IP, many vendors and VMS platforms, cloud or local storage, 7–15+ days' retention — at sites up to ~1,000 km apart. The State wants one platform that reads plates, correlates them continuously with watchlists from VAHAN, SARTHI, eGujCop, AFIS and NAFIS, raises real-time alerts and traces vehicles, growing to ~80,000 cameras on existing infrastructure. VISWAS (7,000+ cameras in Phase I, 10,500+ in Phase II) and 34 Netram centres already exist; the missing piece is the connective layer.

Declared model — Hybrid. **Model 1** (registry and GIS) is the foundation and the system of record for every camera. **Model 3** (federation) is the spine: adapters normalise each vendor's system into one camera contract and one event contract. **Model 2** (direct connection) is used per camera or recorder, over RTSP, HLS, ONVIF or a vendor API, without changing the department's system. PRAHARI delivers **Model 4's analytics** — ANPR, watchlist correlation, cross-camera tracking — but runs them at the edge, so it does not need Model 4's central video backbone, petabyte storage or GPU farm. Full centralisation was evaluated and rejected on bandwidth, storage, compute, cost and data-protection grounds (§3).

Core principle: move observations, not video. Video stays in the department that owns it and is recorded where it is recorded today. A PRAHARI node next to the cameras decodes and analyses the streams and sends only small structured events — a plate read is a few hundred bytes, an evidence crop about 10–30 KB — outbound to the State. Raw video crosses the network only when an authorised officer asks for a specific clip for a specific case.

160 → < 0.5 Gbps

Statewide backbone at peak: all 80,000 streams centralised at 2 Mbps vs PRAHARI's events and crops (~350x; 0.21 Gbps average) **derived** §3, §12

148 / 148

Automated tests passing (ingest, ANPR, grammar, dedup, registry, alerts, trace, audit) **measured** 25 Sep 2026

4 of 4 · 6 / 0

Practice run: stolen vehicle traced across exactly its 4 scripted cameras, in order; 6 correct alerts, 0 false alerts **measured** §16

12 / 12 live

Mixed H.264/H.265, 3 resolutions, 10–25 fps replica grid; every stream recovered after feed restarts (36 reconnects), 0 decode errors **measured** §5.6

~17 ms / frame

Plate detection + reading, one plate, 960–1280 px frame, Apple M1 laptop (p90 24–30 ms) **measured** §8.4

32% → 83%

Watchlist recall at 10% OCR character error: exact match vs PRAHARI incl. operator review; 0 false alerts (synthetic benchmark) **measured** §7.6

What is built and demonstrated today

- **Ingest spine** **BUILT** — catalogue-driven onboarding, RTSP over TCP, PTS timing, measured frame rate, reconnect with backoff, loop handling, mixed H.264/H.265.
- **ANPR for Indian plates** **BUILT** — ~11 MB ONNX models, two-line handling, grammar-constrained decoding, per-pass de-duplication, per-camera capability profiling.
- **Registry and GIS** **BUILT** — three onboarding paths, health map, coverage-gap report, plates CSV report.
- **Watchlist, alerts, trace** **BUILT** — confusion-space matching (GJ01AB1Z34 still matches GJ01AB1234), alert vs review, live push, acknowledgement, route reconstruction with plausibility checks.
- **Accountability** **BUILT** — purpose and case ID required for movement queries; SHA-256 hash-chained audit log.
- **Console and API** **BUILT** — eight browser pages; 29 REST endpoints with OpenAPI documentation.

Cost. For 80,000 cameras the sizing model estimates PRAHARI at **₹49 Cr capex and ₹18 Cr a year opex** — **≈ ₹130 Cr over five years**, against **≈ ₹510–640 Cr** for a centralised design running the same analytics (infrastructure only; **≈ ₹870 Cr** with commercial licences; ±30%, ex-GST) **derived** §12.

Not yet built, and stated as such throughout: operator login and role-based access (today the operator names themselves), the node's outbound mTLS link, disk store-and-forward and bandwidth ladder, SHA-256 evidence manifests, government-database adapters, face recognition, appearance-based re-identification and non-ANPR analytics. Plate accuracy is so far measured on synthetic plates; two-wheeler plates are read poorly. §16.5 lists every limitation; §15 schedules each.

2 Problem understanding and constraints

2.1 The situation as stated, and what follows from it

Fact (source: Problem Statement unless noted)	Detail	Design consequence
26 departments, independent systems	Each owns its cameras, VMS, AMC contract, budget line and liability. Examples named: Home (public spaces, traffic, law and order), Food & Civil Supplies (godowns, PDS shops), RTO (offices, test tracks, checkpoints). The sandbox adds Health, Panchayat, GSRTC and Municipal cameras.	Integration must not require any department to replace, reconfigure or hand over its system. Joining should cost a department nothing and give it something (health monitoring, its own map and gap report).
Heterogeneous infrastructure	Analog and IP cameras; multiple vendors and VMS platforms; H.264 and H.265; varied resolutions and frame rates; cloud and local storage.	One normalised camera contract (CameraInfo) and one event contract (PlateEvent) behind vendor adapters. Never assume a fixed resolution, codec or frame rate.
Retention 7–15+ days, varies	Video is already recorded, locally or in a department cloud.	Do not duplicate video storage centrally. Pull a clip from the owner's recorder on demand, within its retention window.
Geographic dispersion ~1,000 km	Border districts, Valsad, Dahod, Somnath, Jamnagar, Dwarka. Uplinks vary from fibre to thin or intermittent links.	Process at the edge; send events, not streams; keep working through outages and back-fill afterwards.
Scale ~80,000 cameras	Target size of the unified platform.	Horizontal scaling units (one node per site; partitioned central services). Onboarding must be automated — 80,000 cameras cannot be surveyed by hand.
Existing State assets (public sources, Appendix D)	VISWAS Phase I: 7,000+ cameras at ~1,200 junctions, 34 district HQs, pilgrimage centres; Phase II: 10,500+ cameras in Surat, Vadodara, 52 municipalities and inter-state points. 34 Netram district command and control centres.	Netram centres become regional tiers; VISWAS cameras are onboarded as they are, through their existing backhaul. PRAHARI is the connective layer, not a replacement.
Government databases	VAHAN, SARTHI, eGujCop (CCTNS), AFIS, NAFIS: stolen vehicles, wanted persons, missing persons, unidentified bodies, fingerprints.	Watchlist adapters with a stable internal schema. Treat production access as a provisioning step owned by each database's custodian.
Private cameras “wherever feasible and permitted”	Societies, malls, commercial establishments.	Consent tiers per camera with owner revocation; default is “registered”, not live ingestion (§10.9).
Legal envelope (Appendix D)	Digital Personal Data Protection Act 2023; <i>K.S. Puttaswamy v. Union of India</i> (2017) proportionality test for State intrusion into privacy.	Purpose binding, minimisation (no central video), access audit, retention limits, face recognition off by default.

2.2 What the Sentinel sandbox teaches about real feeds

The organisers' sandbox notes describe behaviours any production system meets at scale, and they are treated here as engineering constraints: the catalogue `GET /api/ingest` is the contract while identifiers and URL patterns may change; RTSP must run over TCP; a buffered GOP is replayed faster than real time on connect, so timing must come from stream timestamps (PTS), not arrival; declared frame rates are unreliable and frame intervals uneven; feeds restart, decoder warnings at join are normal, each feed loops with a hard scene cut, and codecs and resolutions are mixed; clients must consume only and open only what they process. Section 5.6 shows how each is handled and tested.

2.3 Non-functional requirements adopted

Quality	Requirement PRAHARI sets for itself
Non-intrusive	No change to any department's cameras, recorders, VMS or firewall inbound rules. PRAHARI is an additional read-only stream client.
Vendor-neutral, open	Open protocols (RTSP, HLS, ONVIF), open model formats (ONNX), documented REST/OpenAPI, adapters as the only vendor-specific code — matching the “Architecture Principles” of Step 3.
Degrade, never fail	One bad camera, one bad catalogue fetch or one bad frame never stops the rest. Loss of the State link does not stop detection at the edge.
Bounded resources	No unbounded queues; one decoder and one latest frame per camera in memory; raw video never stored by PRAHARI; evidence storage capped.
Accountable	Every query on a person's or vehicle's movement is purpose-bound and audited; audit log is tamper-evident.
Measured, not assumed	Camera properties (frame rate, codec, resolution, plate legibility) are measured by the system and written to the registry.

2.4 Assumptions register

These assumptions underpin the design. Each has an owner who can confirm or correct it during the pilot (§15).

ID	Assumption	If wrong
A1	Every camera is reachable as a stream from a host on the owner's network: directly (IP camera) or through its DVR/NVR/encoder/VMS (analog cameras).	Analog cameras without any encoder need a low-cost encoder; listed as a prerequisite in §14.
A2	Each site, or its district Netram centre, can host a node (VM, spare PC or small edge box) with outbound HTTPS to the State data centre.	Stream is relayed to the nearest Netram-hosted node over existing links (§11).
A3	Recorders expose read-only credentials and, ideally, a sub-stream; PRAHARI never needs admin rights.	Department creates a read-only user; §14 lists it as a prerequisite.
A4	Site clocks can be synchronised (NTP) to within a few seconds, or skew can be measured.	Trace flags implausible hops rather than hiding them (§8.6); clock-skew measurement is on the roadmap.
A5	Government databases can supply watchlist extracts or lookups through an authorised interface once the custodian approves.	Scheduled CSV/API import (already built) remains the integration path.
A6	Roughly a quarter to a third of all cameras face traffic in a way that permits plate reading; the rest are corridors, gates, counters and yards.	Capability profiling measures this per camera; sizing (§12) is recalculated from the measured share.
A7	Per-stream bitrates for the Model 4 comparison: 2 Mbps (1080p H.264, conservative); 4 Mbps realistic.	Model 4 figures in §3 scale linearly with the bitrate.

3 Choice of model and justification

3.1 The reference models against the constraints

Model 1 is mandatory and forms the foundation in every option. The real choice is how video and analytics are combined with it. The table compares the options against the constraints of Section 2.

Criterion	Model 2 — direct connect	Model 3 — federation	Model 4 — central VMS & AI	PRAHARI hybrid (1 + 3 + 2, edge analytics)
Continuous backbone load	High if every stream is viewed or analysed centrally	Low (metadata) unless video is relayed	~160 Gbps sustained for 80,000 streams at 2 Mbps	0.21 Gbps average, 0.45 Gbps peak (§3.3, §12)
Central video storage	None or partial	None	~51.8 PB for 30 days, before replication and DR	None; evidence crops and case clips only
Inference	Central GPUs for analytics	Depends on each VMS's analytics	~2,000 GPUs as usually specified; ~224 L4 GPUs for the same 1 fps analytics; plus ~445 recording servers	166 L4 GPUs at the 34 Netram centres + ~3,450 small edge nodes; central tier on CPUs
Department control	Kept, but many inbound connections	Kept	Surrendered to the central platform	Kept; node is outbound-only and removable by the department
Works through WAN outage	No	Partly (local VMS)	No	Yes: detection continues, events back-fill (§6.5)
Heterogeneity	Must speak every protocol centrally	Adapters per VMS	Re-ingest everything into one VMS	Adapters per camera/recorder/VMS at the edge
Data protection (DPDP, Puttaswamy)	Central access to all live video	Moderate	Central retention of all video of all public spaces	Minimised: no central video; purpose-bound, audited queries
Time to statewide	Medium	Medium (per-vendor work)	Longest (network and data-centre build)	Shortest: one software node per site, onboarding automated

3.2 Why full centralisation was considered and rejected

Model 4 is the most complete on paper and was evaluated seriously. For a statewide primary architecture it fails on arithmetic before it reaches policy:

Quantity	Calculation	Result	Label
Backbone, conservative	80,000 cameras × 2 Mbps (1080p H.264)	160 Gbps	derived A7
Backbone, realistic	80,000 × 4 Mbps	320 Gbps	derived A7
Backbone, sub-streams only	80,000 × 512 kbps	40 Gbps	derived
Storage per camera-day	2 Mbps × 86,400 s ÷ 8	21.6 GB	derived
Central storage, 30 days	21.6 GB × 80,000 × 30	51.8 PB	derived
Central GPUs, as usually specified	80,000 streams ÷ 40 streams per GPU at full video rate	~2,000	assumed 40/GPU

Quantity	Calculation	Result	Label
Central GPUs, same analytics as PRAHARI	1 fps ANPR + 0.2 fps light analytics, decode-bound (§12)	~224 L4	derived

The case does not rest on GPUs. A central design running only PRAHARI's sampled analytics needs about as many GPUs as PRAHARI itself places at the Netram centres. It rests on the **WAN** — 160 Gbps, always on, across ~1,000 km and 33 districts, a recurring cost of about ₹60 Cr a year in the sizing model — and on **~52 PB of video storage**, before a DR copy. Section 12 prices both architectures: over five years, ~ ₹130 Cr for PRAHARI against ~ ₹510 Cr for the most favourable central design and ~ ₹640 Cr as usually specified (infrastructure only), or ~ ₹870 Cr with commercial licences. Three further reasons stand independently of cost:

- **Consent.** Model 4 asks 26 departments to give up operational control of systems they own and are liable for. That is a multi-year negotiation. PRAHARI asks each department only to permit a read-only client on its own network.
- **Resilience.** A central-only system goes blind wherever the WAN fails, which is most likely exactly at the remote border and coastal sites the State most wants to watch.
- **Proportionality.** Continuously moving and retaining video of all public spaces in one place creates a capability far beyond what investigations need. Under the *Puttaswamy* proportionality test and the purpose-limitation and minimisation principles of the DPDP Act 2023, the less intrusive design that meets the same operational need is the defensible one.

Model 4 is not discarded entirely: its *analytics* — ANPR, statewide vehicle tracking, watchlist integration — are all delivered, at the edge. Its central-recording function remains available per case, as an on-demand clip pull from the owner's recorder (§6.6).

3.3 What moving observations instead of video means in numbers

A plate read is a structured record of a few hundred bytes (547 B measured, including five ranked candidates); an evidence crop of plate and vehicle is ~5–30 KB. A first-principles estimate — 30% of cameras facing traffic, 0.2 vehicles per second each around the clock, 200-byte events, 10 KB crops assumed — gives:

Flow	Volume per day	Average rate	Label
Events only (24,000 cameras × 0.2 veh/s × 200 B)	~83 GB	~7.7 Mbps	derived
Events + one evidence crop each	~4.2 TB	~392 Mbps	derived
Raw video, all 80,000 cameras at 2 Mbps	~1,730 TB	160,000 Mbps	derived

These first-principles rows are shown only for transparency and are **superseded**: the bottom-up sizing model (§12) corrects their inputs in both directions: fewer plate events (realistic traffic per camera type and a 29% ANPR-capable share), larger crops (15 KB), plus telemetry, clip pulls and protocol overhead, and it provisions for the peak hour rather than the average. Its base case is **0.21 Gbps average and 0.45 Gbps at peak — about 350× less at peak (750× on averages)** derived. Links must be bought for the peak, so the headline used in this document is **160 Gbps → under 0.5 Gbps at peak (~350×)**. Across every sensitivity case in §12 the ratio stays between 110× and 5,500×.

3.4 What each reference model contributes

Model	What PRAHARI takes from it	Where
Model 1 — Registry & GIS	The system of record for every camera: location, department, ownership, stream endpoints, health, and <i>measured</i> capability. Three onboarding paths, map layers, gap analysis, audit trail. Mandatory, and the foundation of everything else.	§5.4, §8.5, §16; prahari/registry/
Model 3 — Federation	Adapter architecture that normalises vendors into one camera contract; event contract and event bus; cross-system correlation (watchlist, trace) over events from all departments; unified alert dashboard.	§4.3, §5.2, §7; prahari/common/contracts.py
Model 2 — Direct connect	Per-camera or per-recorder stream access over RTSP/HLS/ONVIF/vendor API, without middleware in the video path; ANPR metadata, camera-wise indexing, searchable movement records, video wall. Departmental systems unaffected.	§5, §6; prahari/node/
Model 4 — Central AI	The analytics (ANPR, statewide tracking, database integration), tiered storage for events and evidence, DR, encryption, RBAC. Not the central ingestion and recording of all video.	§7, §8, §10, §13

3.5 Precedent

The pattern is established elsewhere and is adapted, not invented. The UK National ANPR Service federates plate reads — not video — from all police forces, at a reported ~90 million reads per day, through a gateway → ingest → screen → exploit pipeline whose shape PRAHARI's central tier follows; it relies on standardised UK plates, which is why PRAHARI replaces exact matching with grammar-constrained, confusion-aware matching for Indian plates (§7). Edge appliances that dial out from existing recorders to a real-time crime centre (e.g. Fūsus in the United States) show that unifying public and private cameras does not require replacing them; PRAHARI keeps the pattern but makes the node software-only and tolerant of poor connectivity. South Korea's district-level integrated CCTV control centres paired consolidation with a statutory privacy framework; PRAHARI pairs integration with DPDP-aligned controls by design. Sources are in Appendix D.

4 Overall architecture

4.1 Architecture principles

- Video stays home; observations travel.** Streams are decoded and analysed next to the cameras. Only events, crops and health reports cross the WAN.
- Department systems are untouched.** PRAHARI is one more read-only stream client. Recording, retention and the existing VMS continue unchanged.
- Outbound-only nodes.** A node opens connections to the State; the State never connects into a department network. **DESIGNED**
- Contracts first.** Three data contracts — `CameraInfo`, `FrameSample`, `PlateEvent` (Appendix B) — separate vendors from analytics and analytics from central services. **BUILT**
- Nothing hardcoded.** The camera set, identifiers and URLs come from a catalogue or the registry, re-read every 60 s. **BUILT**
- Measure, don't assume.** Frame rate, codec, resolution, health and plate legibility are measured per camera and stored in the registry. **BUILT**
- Degrade, never fail.** Every per-camera, per-frame and per-fetch failure is contained and counted. **BUILT**
- Accountability by construction.** Queries on movement require purpose and case; every one is recorded in a tamper-evident log. **BUILT**
- Never silently empty; never hide evidence.** Weak matches go to human review, trace returns ranked near-misses instead of “no result”, and physically impossible hops are flagged, not dropped (they may reveal a cloned plate). **BUILT**

4.2 Statewide tiers: edge, regional, central

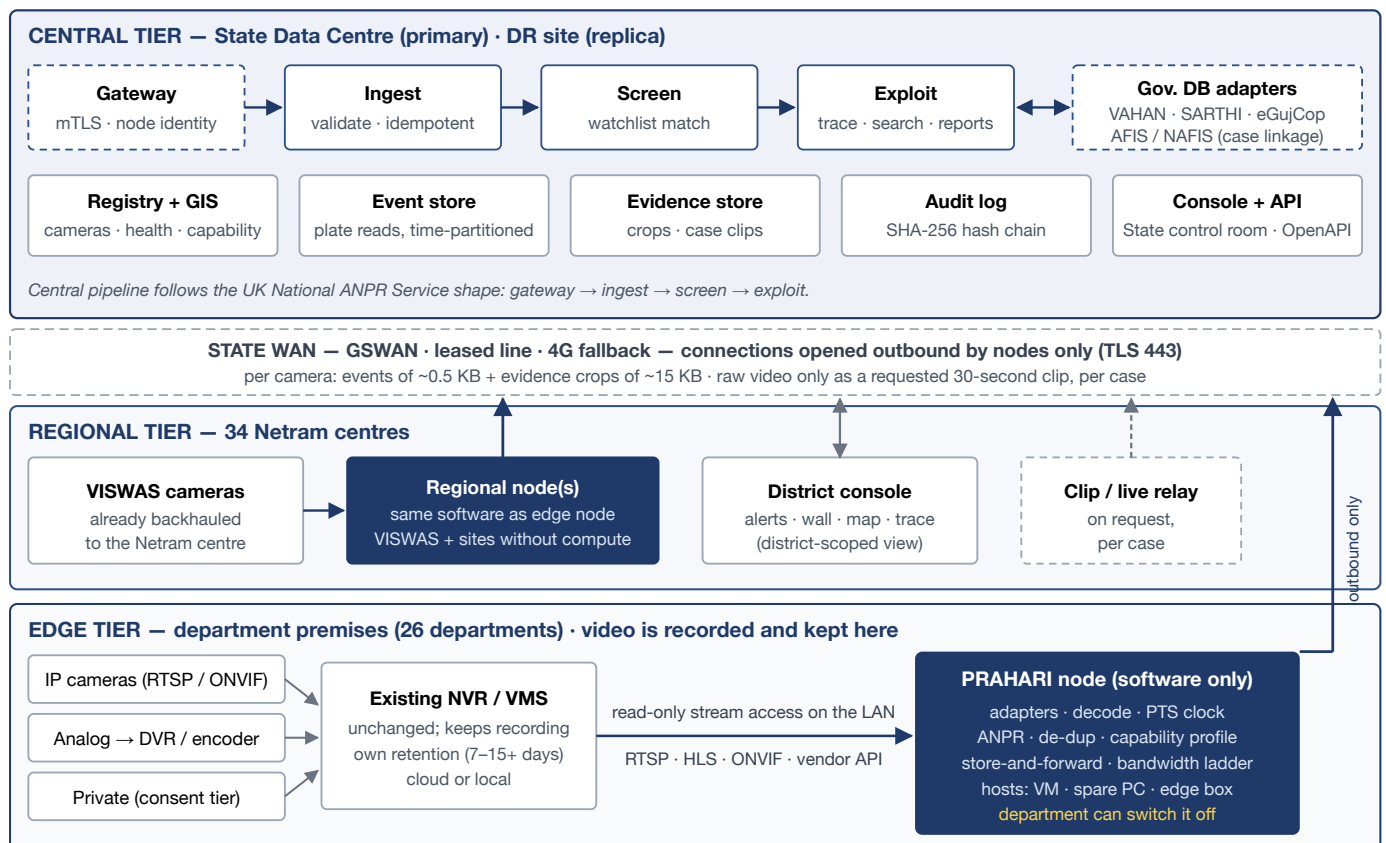


Figure 4.1 — Statewide tiers. Solid outlines are built in the reference implementation (where the tiers run in one process); dashed outlines are designed. The regional tier reuses the 34 Netram centres: VISWAS cameras, already backhauled there, are processed by regional nodes, and district officers use a district-scoped console.

4.3 Component view

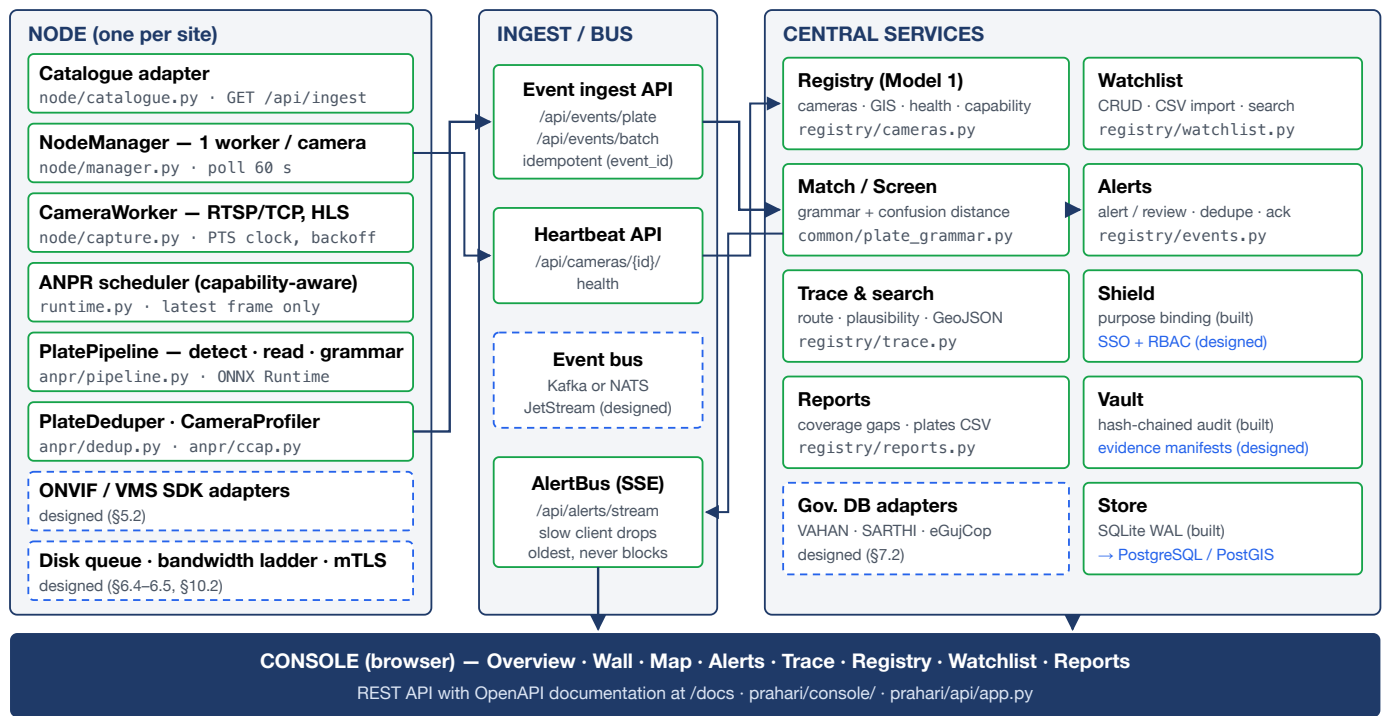
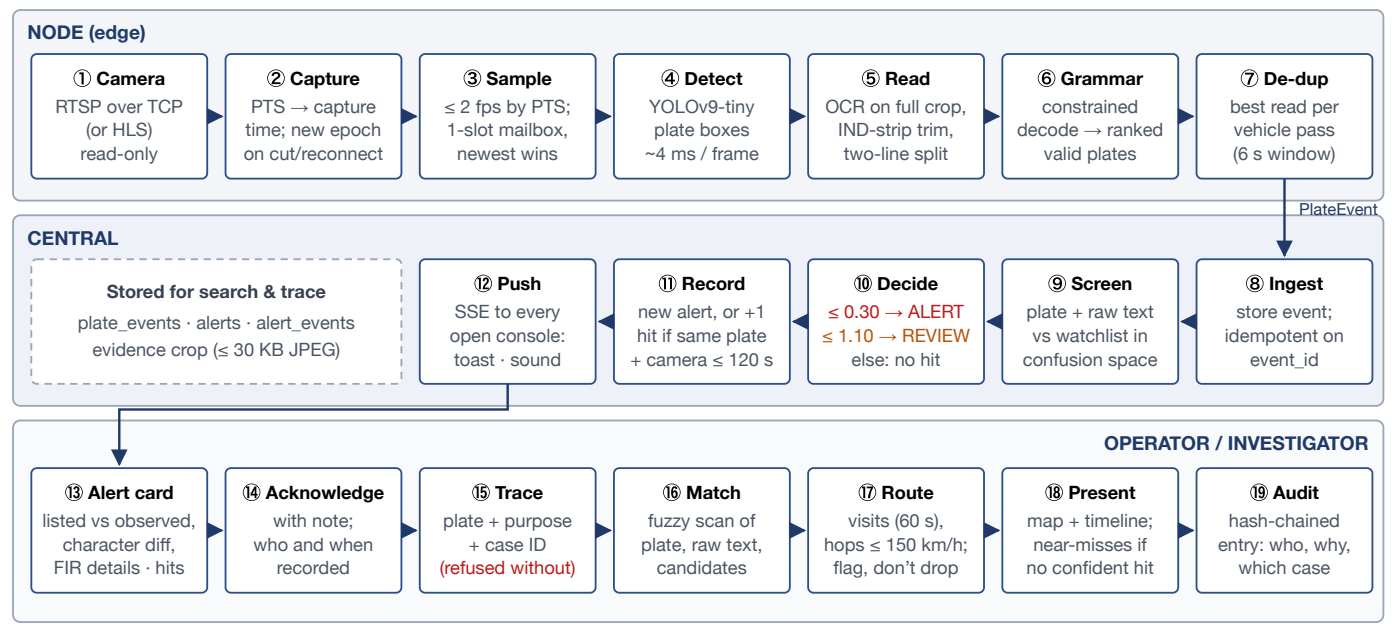


Figure 4.2 — Component view. Green outline: built (file named); dashed blue: designed. In the reference implementation all three groups run in one process (run.py); the arrows are the same interfaces that cross the WAN in production (Table 4.2).

Table 4.2 — Component interactions

From → to	Interface	Payload and semantics	Status
Catalogue / VMS → node	HTTP GET /api/ingest; adapters	Camera list → CameraInfo; tolerant parsing; raw entry kept; re-pollled every 60 s	BUILT
Camera / recorder → node	RTSP over TCP; HLS	Compressed video, read-only, LAN only; decoded in memory, never stored	BUILT
Capture → ANPR	In-process 1-slot mailbox	FrameSample (PTS-derived capture time, epoch); newest frame wins, nothing queues	BUILT
Node → central (events)	POST /api/events/plate, /batch	PlateEvent JSON; idempotent on event_id, so at-least-once delivery is safe	BUILT (in-process call today)
Node → central (health)	POST /api/cameras/{id}/health	State, measured fps, codec, resolution, reconnects, decoder warnings, ANPR profile; every 10 s	BUILT
Ingest → screen → alerts	In-process; production via event bus	Screen each event on corrected plate and raw text; alert/review; 120 s de-duplication	BUILT / bus DESIGNED
Alerts → console	Server-Sent Events /api/alerts/stream	One JSON alert per message; replay of open alerts on connect; 15 s keep-alive	BUILT
Console → trace/search	GET /api/trace/{plate}, /api/events	Requires purpose + case_id; returns sightings, hops, GeoJSON, near-misses; audited	BUILT
Gov. DB → watchlist	Adapter (pull delta / lookup)	Watchlist entries with category and case details; enrichment on hit	CSV/API BUILT ; adapters DESIGNED
Central → node (clip)	Request over the node's outbound channel	“30 s around time T on camera C for case K”; node fetches from the recorder, hashes, uploads	DESIGNED

4.4 End-to-end data flow: plate read → alert → trace



Steps ①–⑦ run at the edge, ⑧–⑫ centrally, ⑬–⑲ in the console. All are built; in production the ⑦ → ⑧ hop crosses the State WAN through the gateway (§6).

Figure 4.3 — End-to-end flow. Thresholds and windows are the defaults in the code: alert ≤ 0.30 and review ≤ 1.10 confusion distance (plate_grammar.py), alert de-duplication 120 s (events.py), per-pass de-duplication 6 s (runtime.py), trace visit collapse 60 s and plausibility 150 km/h (trace.py).

4.5 Reference implementation and production target

The submitted reference implementation runs every tier in one Python process so that it can be installed and demonstrated on a single laptop with no internet access (python run.py). The production target keeps the same components and contracts but deploys them as separate, horizontally scalable services. The suggested technology stacks in the Problem Statement are used where they fit.

Layer	Reference implementation (built)	Production target (designed)
Packaging	One process: run.py starts node, API and console; practice mode includes its own RTSP grid	Node: one OCI container or appliance image per site. Central: containers on Kubernetes across two halls, plus DR
Stream I/O	PyAV (FFmpeg) — RTSP over TCP, RTSPS, HLS; per-camera thread	Same, plus ONVIF discovery and vendor adapters; hardware decode (NVDEC, Quick Sync) where present
Inference	ONNX Runtime; CoreML (Apple), CUDA (NVIDIA, with onnxruntime-gpu) or CPU chosen automatically	Same ONNX models; TensorRT or OpenVINO builds for edge accelerators; batched inference on regional GPU nodes
Event transport	In-process call to the ingest function; HTTP ingest and batch endpoints exposed	Node → gateway over mTLS (HTTP/2 batches); central Kafka (or NATS JetStream) topics partitioned by camera
Storage	SQLite in WAL mode (registry/db.py); evidence JPEGs on local disk with a size cap	PostgreSQL + PostGIS (registry, GIS), TimescaleDB hypertables (events), S3-compatible object store (MinIO/Ceph) for evidence
Live alerts	In-process pub/sub → Server-Sent Events	Same SSE interface, fanned out across API replicas through the bus
Console	Static HTML/JS (no build step), Leaflet maps, snapshot video wall refreshed every ~2 s	Same console; WebRTC/HLS relay for on-demand live view of selected cameras
Identity	Operator identity supplied in an X-Actor header; server binds to 127.0.0.1 by default	State SSO (OIDC) with RBAC and department scoping; mTLS certificates per node

5 Integrating heterogeneous cameras, NVRs and VMS

Heterogeneity is solved once, at the edge, by adapters that turn whatever a department has into two normalised things: a CameraInfo record (who and where the camera is, and how to open it) and a stream of FrameSamples (decoded pictures with trustworthy timestamps). Nothing downstream of the adapters knows which vendor, codec or recorder a camera belongs to.

5.1 Integration surfaces

Source	How PRAHARI connects	Notes	Status
IP camera	RTSP (forced over TCP) or RTSPS, main or sub-stream	Read-only credentials; the camera keeps recording to its own NVR. Credentials are stripped from every log line and status report.	BUILT
Analog camera	Through the DVR, hybrid NVR or video encoder that already digitises it, which exposes RTSP per channel	No change to the analog plant. Where no encoder exists, a low-cost encoder is a site prerequisite (§14).	BUILT (via RTSP)
NVR / DVR channel	RTSP per channel (most NVRs); vendor HTTP API for channel lists	Channel list becomes the catalogue; one worker per channel.	RTSP BUILT ; API DESIGNED
HTTP streaming gateways	HLS (.m3u8) as a fallback where RTSP ports are blocked	Same decoder path; timing still from PTS.	BUILT
WebRTC (WHEP) endpoints	URL recognised and stored in the registry	Used for low-latency live view in the console; not for analytics.	Parsed BUILT ; playback DESIGNED
ONVIF devices	ONVIF Profile S/T: discovery, GetProfiles, GetStreamUri, device time	Fills the catalogue automatically on a department LAN; also reads device clock for skew.	DESIGNED
Departmental VMS	VMS adapter using the vendor's documented SDK or REST API: camera list, live stream URL, playback/export for clip pull, events	One adapter per VMS product, reused across every department that runs it. Adapter certification test suite with record/replay fixtures.	DESIGNED
Cloud-recorded estates	Vendor cloud API or RTSP relay offered by the provider	Treated as a VMS adapter; bandwidth is the provider's egress, so sub-streams preferred.	DESIGNED
Catalogue services	HTTP JSON catalogue such as the Sentinel GET /api/ingest	Tolerant parser; the proven onboarding path for the evaluation grid.	BUILT

5.2 Adapter architecture

Every adapter implements the same small contract, so a new vendor is a new adapter — not a change to analytics, registry or console.

Adapter method	Returns
<code>list_cameras()</code>	list of <code>CameraInfo</code> (id, name, department, lat/lon, codec, resolution, declared fps, live flag, RTSP/HLS/WHEP URLs, raw record)
<code>open(camera)</code>	stream URL or handle the capture worker can decode
<code>health()</code>	adapter-level status (reachable, authorised, last error)
<code>export(camera, t0, t1)</code>	clip from the recorder for a case DESIGNED

Built today. The catalogue adapter (`prahari/node/catalogue.py`) implements `list_cameras()` for any JSON catalogue, and the capture worker (`prahari/node/capture.py`) opens any RTSP, RTSPS or HLS URL through FFmpeg (PyAV). Because most NVRs, encoders and VMS products can publish RTSP, these two cover the majority of sources without vendor code.

Designed. ONVIF and per-VMS adapters implement the same methods. Each ships with a conformance test that replays recorded responses from the real product, so an adapter can be certified without access to the department's live system.

5.3 Catalogue-driven onboarding

For the evaluation grid, the Sentinel catalogue GET /api/ingest is the only source of truth; camera identifiers, the set of cameras and URL patterns can change. The node therefore hardcodes nothing and re-reads the catalogue every 60 seconds **BUILT**:

- **Tolerant parsing.** A bare list, or a list wrapped under `cameras`, `streams`, `items`, `data` and similar keys, or an object keyed by camera id. Field aliases for id, name, department, coordinates (flat, nested under `location/geo`, or GeoJSON `[lon, lat]`), resolution (`1280x720`, `720p`, `[w, h]`), frame rate (`25` or `"25/1"`) and live status. As a last resort, any string that looks like an `rtsp://`, `.m3u8` or WHEP URL is used. The raw entry is always kept verbatim.
- **Safe reconciliation.** New live cameras get a worker; cameras that disappear or go not-live are stopped; a changed URL restarts the worker. A failed catalogue fetch keeps the current camera set, and an empty catalogue must be seen twice in a row before cameras are stopped — a catalogue blip cannot take the wall down.
- **In the registry,** cameras that vanish from the catalogue are marked `absent`, never silently deleted, so their history and evidence remain.

5.4 Three onboarding paths into the registry (Model 1)

Path	Endpoint	Use	Status
1. Catalogue / API sync	POST /api/cameras/sync (URL or inline payload)	Machine onboarding of a whole estate from a catalogue, VMS export or another system	BUILT

Path	Endpoint	Use	Status
2. Bulk import	POST /api/cameras/import (CSV or JSON body)	Department spreadsheets; same column aliases as the catalogue parser; rows that cannot be parsed are returned with the reason	BUILT
3. Manual entry	POST /api/cameras (one camera, JSON); Registry page form	Single cameras, private cameras, corrections	BUILT
Heartbeat (all paths)	POST /api/cameras/{id}/health	Node reports measured status and capability; merged into the camera record	BUILT

Every registry change is written to the audit log (actor, action, parameters).

5.5 Zero-inbound node and zero-change onboarding DESIGNED

The node is placed on the department's own network, next to its recorder. It reaches cameras over the LAN and opens a single outbound TLS connection (port 443) to the State gateway, authenticated with a per-node certificate. The State never opens a connection into a department network.

What a department must do	What it does not have to do
Provide a host (VM, spare PC or small edge box) on the camera network; allow outbound HTTPS to one State address; create a read-only stream user.	Open an inbound port, obtain a public IP, change its VMS or recording, renegotiate its AMC, send video off-premises, or give up control. The department can stop its node at any time; the registry then shows its cameras as offline, and its own recording is unaffected.

In return each department gets, on day one, what most do not have today: live health monitoring of every camera, its own map and coverage-gap report, and a measured record of which cameras work — useful evidence in AMC discussions.

5.6 Real-feed behaviours and how they are handled

Each behaviour in the Sentinel sandbox notes (§2.2) is handled in `prahari/node/capture.py` and `manager.py` and was exercised against a local replica of the grid: 12 cameras in six profiles (H.264 and H.265 at 640×360, 960×540 and 1280×720, 10–25 fps), served over RTSP-over-TCP only, looping, with GOP replay on join (`sandbox/grid.py`, `sandbox/rtsp_server.py`).

Behaviour	Handling	Evidence
RTSP over UDP corrupts frames across NAT	<code>rtsp_transport=tcp</code> on every RTSP open; the replica server refuses UDP with 461.	All replica runs; unit tests on capture options
GOP replayed faster than real time on join	Capture time = offset + PTS, where the offset is the running minimum of (arrival – PTS) in the current epoch; replayed (late) frames can only raise the sample, so the minimum converges to the true offset.	PtsClock unit tests (<code>tests/test_node_capture.py</code>)
Declared fps is unreliable	Rate is measured from packet PTS over a 10 s window; declared fps is stored but never used for timing.	Replica: measured 9.6–25.0 fps against declared 10–25 measured
Non-uniform frame intervals	A gap is not a disconnect; only read timeout (10 s), end of stream or error ends a session. Sampling is PTS-driven and re-anchors after long gaps rather than bursting.	Unit tests on the rate limiter
Feeds restart	Reconnect with exponential backoff: 2 s doubling to 30 s, ±25% jitter; ladder resets after 20 s of healthy streaming. One camera's failure never affects another.	Feed-restart run: 36 reconnects over 150 s (3 per camera), 12/12 live at end measured
Decoder warnings at join (H.265 “Could not find ref with POC”)	libav logs captured per worker thread and counted by kind; never fatal. Bad packets counted and skipped.	Loop run: 341 decoder warnings counted, 0 decode errors, 0 crashes measured
Loop point = hard scene cut	PTS moving backwards, or jumping ahead of wall time by > 5 s, starts a new <i>epoch</i> ; downstream de-duplication closes its open clusters, so nothing is merged across the cut. Every reconnect is also a new epoch.	Loop run with PTS reset: 24 discontinuities (2 per camera) detected measured
Mixed H.264/H.265, resolutions, rates	One decoder per stream; output rescaled per frame to a maximum width with aspect preserved; no fixed-shape batch anywhere.	All six replica profiles live together
Load pacing; no downloads; consume only	Only processed streams are opened (allow-list, <code>MAX_CAMERAS</code>); frames decoded live from the stream, never downloaded; nothing is ever published to the gateway.	By construction

Replica runs on 24 September 2026, Apple M1 (8 cores, 8 GB): 200 s keyframes-only run (12/12 live, 6.5% of one core, 116 MB resident); 150 s full-decode run at 5 fps with repeated server restarts; 100 s full-decode run with PTS reset at loop points. Summaries in `data/stability-*.json`, produced by `python -m prahari.node.stability`.

6 Ingesting and managing streams from dispersed sites

6.1 Where each kind of work runs

Everything that needs the full video — decoding, sampling, plate detection and reading, de-duplication and capability profiling — runs at the edge (or on a Netram regional node for VISWAS and for sites without local compute), because moving the video would cost ~160 Gbps (§3). Watchlist screening, alert decisions, trace, search, registry, reports and audit run centrally, where events from all departments meet and sensitive watchlists stay in one place; their cost is CPU-only and small per event. Live view and clip retrieval flow edge → regional relay → console, on request and per case only.

6.2 Inside a node

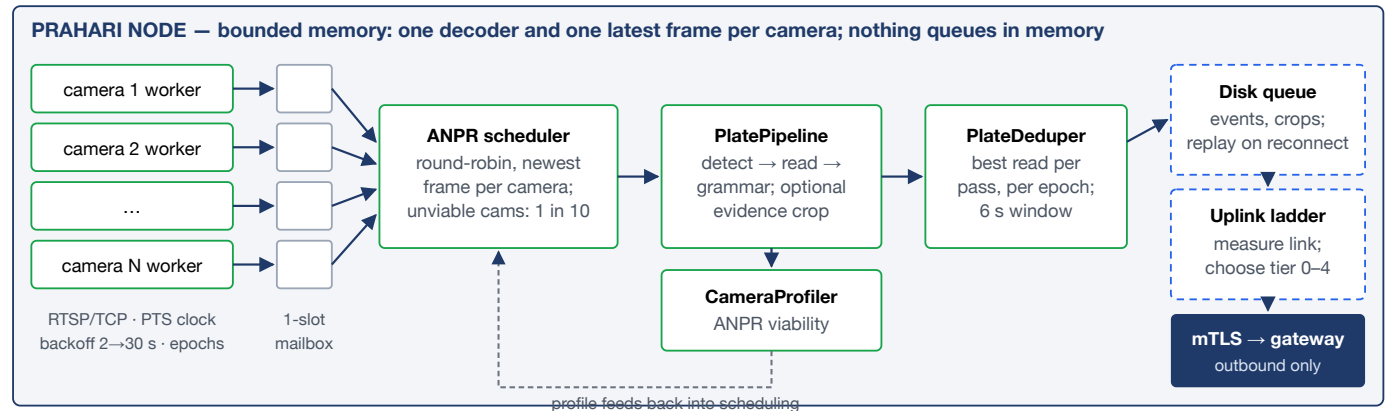


Figure 6.1 — Node internals. Green: built (node/capture.py, node/manager.py, runtime.py, anpr/*). Dashed blue: designed. In the reference implementation the de-duplicated events go straight to central ingest in-process.

- Sampling.** In full-decode mode every frame is decoded (needed for H.264/H.265 reference frames) but only up to TARGET_FPS (default 2) are converted and offered for analysis, selected by PTS. In keyframes mode only keyframes are decoded — about 0.5–1 frame per second at a small fraction of the CPU — for large, low-priority estates. Frames are rescaled to MAX_WIDTH (default 1280 px, because plates need pixels). **BUILT**
- Latest-frame semantics.** Each worker holds only its newest frame. If analysis falls behind, older frames are overwritten, not queued, so latency and memory stay bounded however many cameras are added. **BUILT**
- Capability-aware scheduling.** Cameras that the profiler has measured as unable to read plates (after at least 60 analysed frames) are still sampled, but only one frame in ten, so inference goes where plates are legible. **BUILT**
- Disk discipline.** Raw video is never written. Evidence crops are ≤ 30 KB JPEGs under a hard cap (300 MB by default in the reference build, oldest removed first); all writers stop, loudly, when free disk falls below a floor. **BUILT**

6.3 Measured per-node capacity

Measurement (Apple M1, 8 GB)	Result	Label and source
Ingest only, 12 mixed streams, keyframes mode, 200 s	6.5% of one core; 116 MB RSS	measured 24 Sep 2026, stability run
Ingest only, 12 mixed streams, full decode, 5 fps sampled	~44% of one core; 163–189 MB	measured 24 Sep 2026, stability runs
Ingest only, 48 streams, keyframes mode, 120 s	~49% of one core (steady); ~400 MB (peak 407 MB)	measured 25 Sep 2026; 48/48 live, 0 reconnects, 0 warnings; docs/hld/evidence/stability48-keyframes.json
Ingest only, 48 streams, full decode, 5 fps sampled, 120 s	~168% of one core (steady); ~560 MB (peak 574 MB)	measured 25 Sep 2026; 48/48 live, 0 reconnects, 0 warnings; docs/hld/evidence/stability48-full15.json
ANPR per frame, one plate, 960 px / 1280 px	median 17.3 / 17.4 ms; p90 23.5 / 29.8 ms	measured 25 Sep 2026, prahari.anpr.bench (§8.4)
Streams per ANPR worker at 1 fps, inference only (1000 ÷ p90) — raw, not a planning figure	~40–47	derived ; synthetic frames, decode excluded
Planning figure per laptop-class (M1) node at 1 fps	~14 ANPR streams	derived in §12: ×2 safety factor for real footage, 60% target utilisation
Software decode cost	~20% of a core per 1080p25 stream	derived from the 12-camera full-decode run (111.7 Mpix/s at 43.8% CPU); decode, not inference, is the edge bottleneck, hence hardware decoders in every production node class (§12)

6.4 Low-bandwidth operation: the uplink ladder DESIGNED

A node measures its own uplink continuously and steps down a ladder instead of failing. Watchlist hits are always sent first.

Tier	Measured uplink	What the node sends
4	> 10 Mbps	Events, crops, periodic thumbnails for the wall, live relay on request
3	2–10 Mbps	Events and crops
2	0.2–2 Mbps	Events; crops only for watchlist hits; other crops kept locally and fetched on demand
1	< 200 kbps	Events only, compressed and batched; watchlist hits immediately
0	Offline	Detection continues; everything goes to the disk queue and is replayed on reconnect, oldest first, hits first

The per-site uplink each tier needs, and how long a node can queue offline, are computed in Section 12 for small, medium and large sites.

6.5 Store-and-forward and delivery guarantees

- **Central side** BUILT: ingestion is idempotent on `event_id` (INSERT OR IGNORE), and a batch endpoint exists for flushes. Resending an event after a timeout can never create a duplicate sighting or alert.
- **Node side** DESIGNED: an append-only on-disk queue with a size cap; events are acknowledged by the gateway per batch and only then removed. Eviction under a full disk removes non-hit crops first, then non-hit events oldest-first; watchlist hits are never evicted. On reconnect the backlog is replayed in time order while live traffic continues. Because events carry capture time, back-filled events land in the right place in every trace.

6.6 Viewing, health and clip retrieval

- **Video wall** BUILT: every camera as a tile showing its latest analysed frame with plate boxes drawn, refreshed about every 2 s and only while the tile is visible, with per-tile backoff for failing cameras. Filters by department, state and name.
- **Health** BUILT: every 10 s each camera's state (online, degraded, offline), measured fps, codec, resolution, reconnect count, decoder-warning count, last error and ANPR profile are written to the registry; the map colours cameras by health and the gap report lists offline, stale and never-seen cameras by department.
- **On-demand live view and clip pull** DESIGNED: an authorised user requests a live view or a 30-second clip around a sighting, with a case ID. The request travels down the node's outbound channel; the node reads from the department's recorder (or relays the live stream through the regional relay), hashes the clip at the edge and uploads it to the evidence store. The request, the approver and the hash are audited.

7 Watchlist integration and real-time correlation

7.1 Watchlist entities

The watchlist is a table of plates with a category, a free-form details record (FIR number, police station, vehicle description, notes), a source and an active flag BUILT (`registry/watchlist.py`). Entries are added one at a time, imported as CSV, or set through the API; removal is a soft delete so past alerts stay explainable.

Category (code)	What a plate hit means	Primary source (production)
Stolen vehicle (<code>stolen_vehicle</code>)	The vehicle itself is sought	eGujCop FIRs; inter-state alerts; VAHAN theft flags where provided
Wanted person (<code>wanted_person</code>)	A vehicle known to be used by a wanted person or absconder	eGujCop (wanted / absconders)
Missing person (<code>missing_person</code>)	A vehicle linked to a missing-person or kidnapping case	eGujCop missing-person reports
Blacklisted (<code>blacklisted</code>)	Vehicle barred by an authority (e.g. transport, excise)	RTO / department lists
Suspect (<code>suspect</code>)	Vehicle of interest in an investigation ("be on the look-out")	Investigating officer, time-limited

Persons are matched today through vehicles linked to them. Matching a person by face requires FRS, which is designed but deliberately not enabled (§8.7). Fingerprint records (AFIS/NAFIS) and unidentified-body records cannot be correlated with video; they are linked to a PRAHARI case by case ID for investigators, not screened against camera feeds.

7.2 Government database adapters DESIGNED

Each database is reached through an adapter that maps its records into the watchlist schema or into an enrichment record. The adapters are not implemented in the submitted code; production access is a provisioning step owned by each database's custodian (credentials, network route, a memorandum on permitted use). Until then — and permanently as a fallback — the built CSV/API import accepts

scheduled extracts.

System	What PRAHARI uses it for	Mode	Controls
eGujCop (CCTNS)	Source of stolen-vehicle, wanted and missing-person watchlist entries; alert outcome written back as a case note (if permitted)	Scheduled delta pull (e.g. every 5 min) + on-demand refresh	Entries carry FIR/case reference; expiry when case closes; every sync audited
VAHAN	Enrichment of a hit: registered make, model, colour, RTO — lets the operator check the observed vehicle matches (detects cloned plates); theft flags if provided	On-demand lookup per alert or per trace, not bulk	Lookup only for an open alert or a purpose-bound query; cached briefly; audited
SARTHI	Licence details of a person already linked to a case	On-demand lookup by an authorised investigator	Investigator role and case ID required; audited
AFIS / NAFIS	No video correlation; case linkage only	Reference by case ID	Access stays within the fingerprint bureau's own systems

Adapter interface. `fetch_changes(since) → [WatchlistEntry]`, `lookup(plate) → EnrichmentRecord`, `health()`. Each adapter is tested against recorded request/response fixtures, so it can be developed and certified before live credentials exist.

7.3 Continuous correlation pipeline BUILT

1. **One event per vehicle pass.** A car crossing a camera is read in several frames. The de-duplicator groups reads at one camera that are within confusion distance 1.0 of each other and within a 6 s window (by capture time), and emits the *best* read of the pass (grammar-valid first, then OCR × detection confidence). A reconnect or loop point closes all open groups. A “first read” mode trades read quality for lower latency.
2. **Store** the event (idempotent).
3. **Screen** both the grammar-corrected plate and the raw OCR text against every active watchlist entry, keeping the best distance per entry.
4. **Decide.** Distance $\leq 0.30 \rightarrow$ **alert**. Otherwise distance $\leq 1.10 \rightarrow$ **review** (at most three review items per event). Beyond that, no hit.
5. **De-duplicate alerts.** A new sighting of the same watchlist plate at the same camera within 120 s increments the open alert's hit count instead of creating a new alert, and is linked to it.
6. **Publish** the alert to every connected console (§9).

7.4 Grammar-constrained decoding

An Indian plate is not an arbitrary string. `prahari/common/plate_grammar.py` encodes the structure BUILT:

- **Nine formats**, e.g. SS DD AA DDDD (GJ 01 AB 1234), single-letter and three-letter series, single-digit districts, Delhi-style category letters, older plates without a series, and Bharat-series (DD BH DDDD AA).
- **A closed vocabulary** of 36 current State/UT codes plus 4 historical codes still on the road, and the highest issued RTO district number per State (Gujarat enumerated exactly, GJ-01 to GJ-39, with office names).
- **Beam search** (beam 64, top 5) over positions, where each character may only be replaced by a character of the class the slot requires, and only through a plausible OCR confusion. 6J01A8I234 becomes GJ01AB1234: 6J is not a State code, the sixth character must be a letter, and 6→G, 8→B, I→1 are common confusions. Each candidate carries its edits (“pos0 6→G”), so the correction is explainable.
- The ANPR pipeline additionally tries removing a stray leading I / IN / IND read from the blue HSRP strip, and single-character deletions when a two-line split duplicates a character at the seam.

7.5 Matching in confusion space

Watchlist lookup never uses string equality. Distance is a weighted edit distance in which substituting characters that OCR commonly confuses is cheap and anything else costs 1.0 (as does an insertion or deletion) BUILT:

Pair	O↔0	I↔1	B↔8	S↔5	G↔6	Z↔2	D↔0	Q↔0	L↔1	A↔4	T↔7	other
Cost	0.08	0.08	0.12	0.12	0.18	0.18	0.22	0.25	0.25	0.30	0.30	1.0

With the alert threshold at 0.30, one common confusion — or two of the cheapest — still raises an alert (a misread GJ01AB1Z34 is 0.18 from GJ01AB1234), while a single unrelated character (distance 1.0) goes to human review rather than an automatic alert. The costs are hand-set priors today; they will be re-fitted from the confusion matrix measured on real footage in the pilot.

7.6 Benchmark: exact matching versus PRAHARI matching

Reproducible with `python bench/bench_plate_matching.py` (200-plate watchlist, 600 observations per error rate, seed 7, run 18 Sep 2026) measured.

OCR character error	Exact match	PRAHARI automatic alert	PRAHARI incl. review	False alerts
0%	100.0%	100.0%	100.0%	0.0%
5%	59.9%	73.9%	96.9%	0.0%
10%	32.4%	48.1%	82.7%	0.0%
15%	16.1%	32.4%	69.2%	0.0%
20%	10.9%	21.9%	53.5%	0.0%
30%	2.7%	6.1%	29.9%	0.0%

How to read these numbers

- **Synthetic.** Plates are generated and corrupted in software. The benchmark isolates the matching step from OCR quality; it is not a field result.
- **Not circular.** Corruption is a mixture: 70% attested confusion flips, 12% dropped characters, 10% arbitrary wrong characters, 8% spurious insertions — the last three are errors the confusion table does not anticipate.
- **“Incl. review” is assisted recall.** Those are ranked candidates shown to an operator with evidence, not automatic alerts.
- **Watchlist size 200.** Larger watchlists raise collision pressure. Re-running at 10,000 and 100,000 entries is an open item before the pilot (§15.2).

7.7 Scaling the matcher

The built matcher compares each observation with every active watchlist entry, which is fast for thousands of entries and simple to verify. For statewide watchlists of lakhs of entries the design **DESIGNED** adds candidate generation: (1) exact hash lookup of the observation and of its top grammar-corrected candidates; (2) a blocking index on a confusion-canonical form (O/0, I/1, B/8... folded to one symbol) to fetch near neighbours; (3) the existing weighted distance only on that short list. Screening is partitioned by camera across stateless workers (§13), and the watchlist is replicated to each worker in memory.

8 AI video analytics

8.1 Capability status

Capability	Status	Scope and notes
ANPR — single-line plates	BUILT	Detector + OCR + Indian grammar; strongest case (§8.3)
ANPR — two-line car plates	BUILT	Two-line split hypothesis; needs ~1.5× the pixels of a single-line plate
ANPR — two-wheeler plates	BUILT (weak)	Detector finds 13–18% of synthetic two-wheeler plates; retraining is roadmap item 1
Camera capability profiling (ANPR viability)	BUILT	Per camera, from its own frames; drives scheduling and the gap report (§8.5)
Cross-camera vehicle tracking (plate-anchored trace)	BUILT	Fuzzy matching, visit collapse, per-hop plausibility, GeoJSON route (§8.6)
Watchlist correlation and alerts	BUILT	§7, §9
Appearance re-identification to bridge unreadable plates	DESIGNED	Vehicle embedding + type + colour, gated by travel time (§8.6)
Scene classification and per-camera analytic selection	DESIGNED	Extends the profiler: road, corridor, gate, counter, platform, yard
Facial recognition (FRS)	DESIGNED — off by default	Case-scoped, supervisor-authorised, human-confirmed (§8.7)
Vehicle make / type / colour	ROADMAP	Cross-check against VAHAN to detect cloned plates
Person detection, crowd density, intrusion, loitering	ROADMAP	For non-traffic cameras (hospitals, depots, godowns, offices) selected by scene class (§8.8)

8.2 ANPR pipeline and models

Stage	Model / method	Details
Plate detection	YOLOv9-tiny licence-plate detector, end-to-end ONNX (open-image-models), 384 px input	7.77 MB. Confidence threshold 0.30. Execution provider chosen automatically: CoreML on Apple silicon (~5× faster than CPU for the detector), CUDA on NVIDIA with <code>onnxruntime-gpu</code> , otherwise CPU.

Stage	Model / method	Details
Plate reading	CCT-XS v2 “global” OCR (fast-plate-ocr), ONNX, 128×64 input, 10 character slots, alphabet 0–9 A–Z	3.34 MB. Trained on plates from 65+ countries; India is <i>not</i> among its training regions, so everything Indian (two-line handling, grammar) is added on top.
Read hypotheses	Full crop; crop with the blue “IND” HSRP strip trimmed; two-line split at the row of least ink (30–70% band, 4% overlap) when width ÷ height < 2.6	All hypotheses in one OCR batch; the best under grammar validity × confidence is kept.
Grammar	Constrained decode (§7.4)	Up to 5 ranked valid plates per read, each with its edits.
Evidence	Context crop around the plate, plate boxed, JPEG ≤ 30 KB, ≤ 360 px wide	Linked to the event; served to the console.

Total model weight ~11 MB (`prahari/anpr/models.py`); models are stored locally and the platform runs with no internet connection. Each `PlatePipeline` holds its own ONNX sessions, so a node scales by running one pipeline per worker.

8.3 Measured accuracy (synthetic plates)

`python -m prahari.anpr.evaluate`, 60 scenes per cell, 24 September 2026 **measured**. “Detected” = a detection overlaps the true plate ($\text{IoU} \geq 0.3$); “correct” = the best grammar-corrected plate equals the truth; rates are over all scenes (a missed detection counts as wrong).

A. End-to-end on synthetic scenes

Frame width / layout	Detected	Correct
1280 px · single-line	97%	95%
1280 px · two-line car	80%	65%
1280 px · two-wheeler	18%	17%
960 px · single-line	92%	80%
960 px · two-line car	85%	52%
960 px · two-wheeler	13%	13%
640 px · single-line	95%	43%
640 px · two-line car	78%	18%
640 px · two-wheeler	15%	3%

B. Reading only, by plate height in pixels

Plate height	Single-line	Two-line	Two-line, no split
16 px	3%	0%	0%
20 px	25%	0%	0%
24 px	42%	0%	0%
32 px	97%	25%	5%
48 px	100%	75%	43%
64 px	100%	87%	62%

What this shows. Resolution is the dominant factor: 1280 px frames read single-line plates almost perfectly on synthetic data, 640 px frames do not, which is why the default analysis width is 1280. Plates below ~24 px tall are not readable; the capability profiler uses that as its floor. The two-line split roughly doubles two-line accuracy at 48 px (43% → 75%). Two-wheeler plates fail mainly at *detection*: the detector was trained largely on car plates.

Caveat. These are synthetic, clean, well-lit plates rendered into synthetic scenes (`prahari/anpr/synth.py`). Real CCTV adds night, glare, motion blur, dirt, hand-painted fonts and angle. Field accuracy will be lower and will be measured on the government feeds and in the pilot before any accuracy figure is used operationally.

8.4 Measured throughput

Frame	Median ms	p90 ms	Detector ms	OCR ms	Streams @ 1 fps	Streams @ 2 fps
960 px, no plate	3.6	4.9	3.6	0.0	202	101
960 px, one plate	17.3	23.5	4.1	12.8	42	21
1280 px, no plate	3.8	6.2	3.8	0.0	161	80
1280 px, one plate	17.4	29.8	4.4	12.4	33	16

`python -m prahari.anpr.bench --iters 40`, Apple M1 (8 GB), detector on CoreML, OCR on CPU, 25 September 2026, with other workloads running (load average ~5) **measured**. “Streams” = $\lfloor 1000 \div (\text{p90} \times \text{fps}) \rfloor$ for one worker, inference only **derived**: decoding is extra and is accounted for in the sizing model (§12). Most CCTV frames contain no plate, so the one-plate row is the conservative case. Repeated runs vary with machine load (p90 at 960 px between ~21 and ~25 ms across saved runs); the sizing model (§12) uses its own saved run, `docs/hld/bench-anpr-m1.json`.

8.5 Camera capability profiling **BUILT**

Most CCTV was not installed for ANPR: corridors, gates, counters, wide shots with plates a few pixels tall. Rather than assume, each camera is measured from the frames it actually delivers (`prahari/anpr/ccap.py`, health loop in `runtime.py`):

Measured per camera	Rule / use
Frames analysed, plate detections	At least 30 frames and 5 plates before judging; otherwise “insufficient data” (a park that never sees a plate is not called broken)
Median plate height (rolling 2,000)	≥ 24 px required (from the reading curve in §8.3)
Valid-read rate	Share of raw reads that already parse as Indian plates; ≥ 0.33 required, else “reads illegible” (glare, angle, blur)
Corrected-read rate	Share the grammar can repair; reported
Measured fps, codec, resolution, reconnects, decoder warnings, last error	Stored with the ANPR verdict in the registry’s capability record every 10 s

The verdict drives inference scheduling (§6.2), appears in each camera’s detail view and in the gap report’s “ANPR-unviable” list with the reason, and replaces most of the survey questions departments would otherwise be asked (§14). **Designed extension:** a scene classifier (road, junction, corridor, gate, counter, platform, yard) that selects which analytic each camera runs.

8.6 Cross-camera tracking and route reconstruction

BUILT

Given a plate, GET /api/trace/{plate} (prahari/registry/trace.py) reconstructs the vehicle’s movement:

1. **Fuzzy match.** Every stored reading of every event in the time window — corrected plate, raw OCR text and each ranked candidate — is compared to the query by confusion distance; events within 0.5 are sightings. Distances are computed once per distinct string, which keeps a busy grid fast.
2. **Visits.** Consecutive sightings at the same camera within 60 s collapse into one visit with first/last time, dwell and read count.
3. **Hops.** For each consecutive pair of visits: time gap, straight-line distance and implied speed. Straight-line distance is a lower bound on road distance, so the implied speed is a lower bound on the real speed: if even that exceeds 150 km/h, the hop is impossible. Such hops are **flagged and kept**, with the reason — misread, clock skew or cloned plate — because an investigator needs to see them.
4. **Output.** Ordered sightings with camera name, department, location, time (IST), match distance, confidence and evidence crop; path length; GeoJSON (route line + sighting points) for the map; the grammar verdict on the query with “did you mean” repairs.
5. **Never an unexplained empty result.** With no confident sighting, ranked near-miss readings (with cameras and times) are returned and labelled weak where appropriate.

Trace requires a purpose and a case ID and is written to the audit log (§10.4).

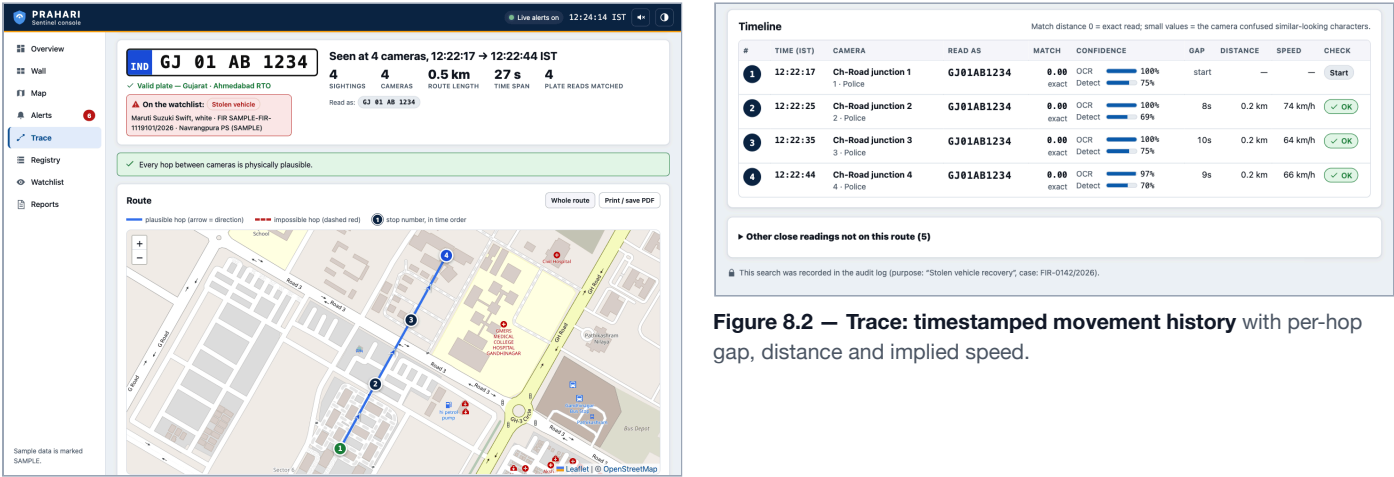


Figure 8.2 — Trace: timestamped movement history with per-hop gap, distance and implied speed.

Figure 8.1 — Trace: route on the map. Practice-mode run; sightings numbered in time order, implausible hops shown in red.

Designed extensions. (a) Road-network travel times (OpenStreetMap graph) instead of straight lines, for tighter plausibility gates. (b) **Re-ID bridging:** when the plate is unreadable at a camera between two confident plate sightings, candidate vehicles there are ranked by appearance embedding, type and colour, and accepted only if the travel times on both sides are feasible; each bridged hop is labelled “Re-ID” with its confidence, never presented as a plate read. (c) Make/colour cross-check against VAHAN to surface cloned plates.

8.7 Facial recognition — proportionate design

DESIGNED, not deployed

FRS is the most intrusive analytic a State can run. PRAHARI’s design treats it as an exception, not a feed-wide default:

- **Off by default** on every camera. Enabled per case, for named cameras and a limited time window, only with recorded authorisation by a supervisory officer designated by the Department; the authorisation, its scope and its expiry are audited.
- **Case-scoped gallery.** 1:N search only against the faces attached to that case (wanted, missing person), never against a population database. Templates are deleted when the authorisation ends unless held as case evidence.

- **Human decision.** A face match is always a *review* item for a trained officer; it never triggers action automatically.
- **Evaluated before use.** Accuracy and demographic error rates measured on representative Indian data; a data-protection impact assessment published before any deployment; periodic independent audit.

8.8 Other analytics

ROADMAP

About two-thirds or more of the State’s cameras watch places where no plate is visible (§2.4, A6). For them, the same node runs different analytics chosen by scene class, emitting new event types on the same contract: person and vehicle detection and counting (depots, platforms), crowd density (pilgrimage centres, hospitals), intrusion and after-hours activity (godowns, PDS shops, offices), loitering. These reuse the detection runtime and the alert workflow; each is validated per scene type before it can raise alerts.

9 Alert generation and notification workflow

9.1 Workflow

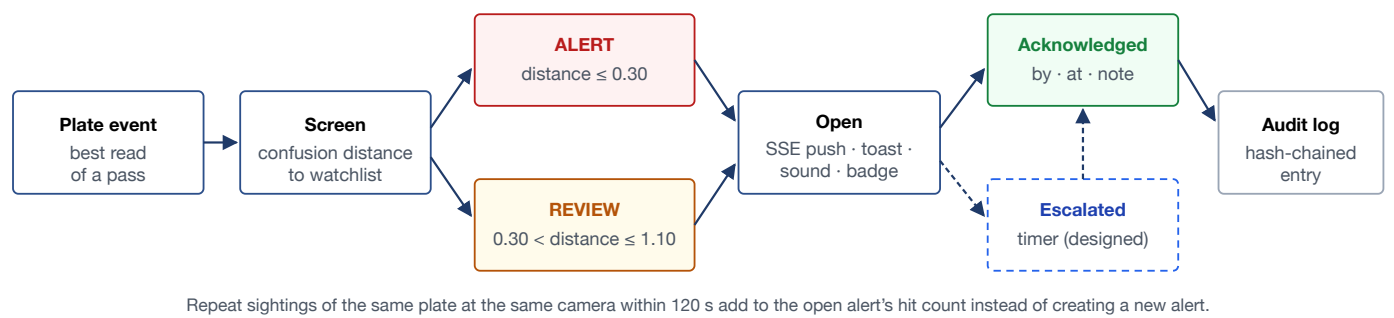


Figure 9.1 — Alert life cycle. Built except escalation (dashed).

9.2 Prioritisation

Built: two classes — *alert* (confident match; red) and *review* (weaker match that a person must confirm; amber) — plus watchlist category, match distance and score on every item; filters by kind and status (open, acknowledged, all); open count on the navigation badge; newest first; items that arrived while the page was open are highlighted.

Designed: a priority matrix combining category and confidence, configurable per department, with acknowledgement targets that drive escalation:

Priority	Condition (default)	Acknowledge within	Escalate to (if not acknowledged)
P1	Alert on stolen vehicle, wanted or missing person	2 minutes	District Netram shift supervisor; then State control room
P2	Alert on blacklisted or suspect vehicle	10 minutes	District supervisor
P3	Review items (any category)	Within shift	Review queue owner

Times are proposals for the Department to set; they are not operational commitments.

9.3 Visualisation and operator interaction

BUILT

- **Alert card:** kind badge; watchlist plate and observed plate side by side with the differing characters highlighted, so the operator sees exactly which characters were corrected; camera name and department; time and age; category and case details (FIR number, police station, vehicle description); how many times it has been seen at that camera. Review items carry the instruction to check the camera picture before acting.
- **Live arrival:** a toast with optional sound, an updated badge, and the card inserted at the top of the Alerts page and in the Overview.
- **Acknowledge with a note** (e.g. “PCR dispatched” or “false positive: GJ01AB1235”); who and when are recorded on the alert and in the audit log. Acknowledged items remain visible under the “Acknowledged” filter.
- **Follow-up:** “Trace this plate” on every card opens the trace page for that plate (purpose and case ID still required).

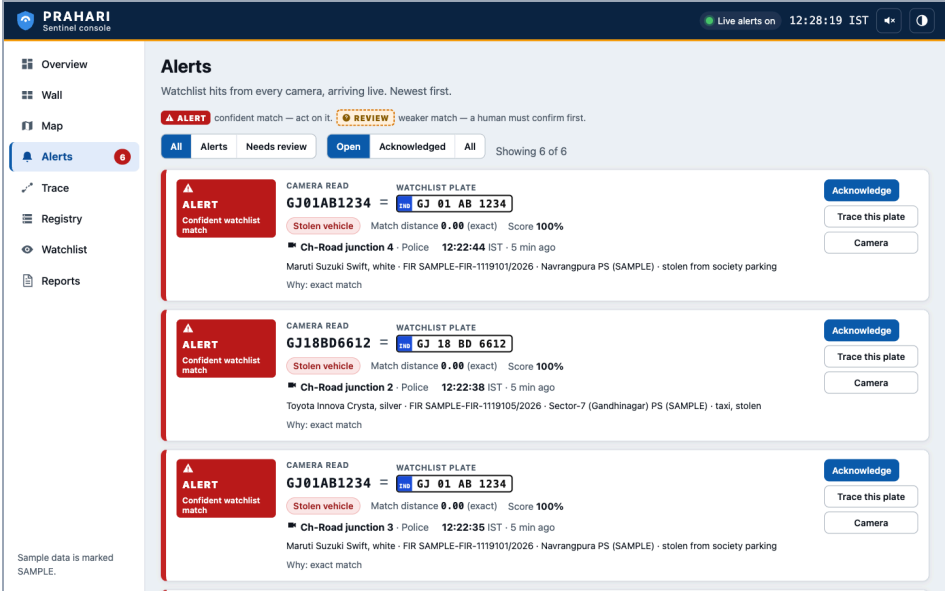


Figure 9.2 — Alerts page (practice mode): confident alerts and review items, with filters and acknowledgement.

9.4 Notification channels

Channel	Behaviour	Status
Console, live	Server-Sent Events at <code>/api/alerts/stream</code> : one JSON alert per message; on connect, optional replay of the N latest open alerts; keep-alive every 15 s; client reconnects after 3 s. A slow client drops its oldest queued items instead of slowing ingestion.	BUILT
API / other systems	GET <code>/api/alerts</code> (filters: status, kind, since); acknowledgement via POST <code>/api/alerts/{id}/ack</code>	BUILT
District console routing	Alerts routed to the Netram centre of the camera's district; State control room sees all	DESIGNED
SMS / mobile push to field units; Dial-112 dispatch	For P1 alerts, through the State's existing messaging and emergency-response systems where the Department permits	DESIGNED
Case write-back	Alert outcome recorded against the case in eGujCop, if permitted	DESIGNED

10 Cybersecurity, privacy and evidence integrity

10.1 Threats and controls

Threat	Control	Status
Compromise of a department network through the integration	Outbound-only nodes; no inbound port, no public IP; the node needs only read-only stream credentials	DESIGNED
Rogue node injecting false events	mTLS with a per-node certificate from a State CA; gateway binds each node to its registered cameras; events for unknown cameras are reported	DESIGNED (unknown-camera report built)
Interception on the WAN	TLS 1.2+ on every hop; no plaintext event or video transport	DESIGNED
Insider trawling movement data	Purpose + case ID required for trace, plate search and watchlist search; every such query audited; RBAC and department scoping	Purpose binding BUILT; RBAC DESIGNED
Tampering with the record of who looked at what	Append-only audit table (database triggers) + SHA-256 hash chain + on-demand verification + external anchoring of the head hash	BUILT (anchoring DESIGNED)
Tampering with evidence	Hash at the edge at capture; signed manifest; write-once storage for case evidence	DESIGNED
Credential leakage	Credentials stripped from URLs in logs and status; sensitive keys redacted in audit parameters; secrets outside code	BUILT (vault DESIGNED)
Resource exhaustion (disk, memory)	Bounded memory per camera; evidence cap; writes stop below a disk floor; slow SSE clients drop oldest	BUILT
Unauthorised console / API access	Today: server binds to 127.0.0.1 by default, operator identity is self-asserted. Production: SSO, RBAC, restricted CORS, rate limits, WAF at the gateway	DESIGNED

10.2 Network architecture and segmentation DESIGNED

- **Zone 1 — camera networks** (department LANs/VLANs): cameras and recorders. Never routed to the WAN by PRAHARI.
- **Zone 2 — node**: two interfaces or a firewall rule set — read-only access into Zone 1 on stream ports; outbound 443 to the State gateway only. No inbound rule.
- **Zone 3 — State WAN** (GSWAN or leased lines): carries only mTLS sessions from nodes and consoles.
- **Zone 4 — central DMZ**: gateway (mTLS termination, rate limiting, WAF) and console front-end.
- **Zone 5 — central services and data**: ingest, screening, trace, databases, object store, audit — reachable only from Zone 4 and the administration network; database adapters to government systems through their prescribed secure channels.
- **Administration** through a bastion with MFA; all administrative actions logged.

10.3 Identity and role-based access DESIGNED

Built today: every write and every sensitive read records an actor, taken from the `X-Actor` header. This is accountability without authentication and is adequate only for a controlled demonstration. Production uses State single sign-on (OIDC) with the following roles, each scoped to departments and districts:

Capability	Control-room operator	Investigator	Supervisor	Dept. admin	Auditor	System admin
Wall, map, camera health	✓	✓	✓	own dept	—	—
View and acknowledge alerts	✓	✓	✓	—	—	—
Trace / plate search (purpose + case)	—	✓	✓	—	—	—
Request clip / live view	—	request	approve	—	—	—
Add / import watchlist entries	—	propose	approve	—	—	—
Authorise FRS for a case	—	—	✓	—	—	—
Onboard / edit cameras	—	—	—	own dept	—	✓
Read and verify audit log	—	—	own actions	—	✓	—
Configuration, keys, deployment	—	—	—	—	—	✓ (no data access)

10.4 Purpose-bound queries BUILT

`GET /api/trace/{plate}`, `GET /api/watchlist/search` and `GET /api/events?plate=...` refuse to run (HTTP 400, `purpose_binding_required`) unless both `purpose` and `case_id` are supplied. The console asks for both before a trace. The query, its parameters, the actor, the purpose, the case and the result count are written to the audit log; plates CSV exports are audited too. This makes every look-up of a person's or vehicle's movement attributable and reviewable (test: `tests/test_registry_audit.py`).

10.5 Tamper-evident audit log BUILT

- Each entry stores time (UTC, microseconds), actor, action, purpose, case ID, parameters (sensitive keys redacted), the previous entry's hash and its own hash = SHA-256 over all of these (`prahari/registry/audit.py`). The first entry links to a genesis value of 64 zeros.
- Database triggers reject any `UPDATE` or `DELETE` on the table; writes use an immediate transaction so the chain stays linear under concurrency.
- `GET /api/audit/verify` recomputes the chain and reports the first entry whose content or link fails, or returns the current head hash. Editing, deleting or re-ordering any entry — even by someone who drops the triggers — is detected.
- **Known limit, stated**: deleting the newest entries leaves a shorter chain that is still consistent. The design closes this by anchoring the head hash externally at fixed intervals (daily report countersigned by the auditor, or a timestamp service) DESIGNED.

10.6 Evidence integrity and chain of custody

Element	Behaviour	Status
Detection record	Every plate event keeps camera, capture time (from PTS), raw OCR text, corrected plate, candidates with edits, confidences, bounding box and evidence-crop path; stored unchanged	BUILT
Evidence crop at capture	Context crop with the plate boxed, written at detection time and linked to the event	BUILT
Hash at the edge	SHA-256 of each crop and clip computed on the node at capture, stored in the event, re-verified on upload and on export	DESIGNED
Signed manifest	Camera, PTS and wall-clock time, node identity, model name and version that produced the detection, operator, case, purpose, hashes; signed with the node's key and countersigned centrally	DESIGNED
Export bundle	Crops/clips + manifest + audit extract + verification tool, verifiable offline by a third party (e.g. a forensic laboratory) without the platform	DESIGNED

Element	Behaviour	Status
Case hold	Evidence linked to a case moves to write-once storage and is exempt from routine deletion	DESIGNED

In the reference build the evidence folder has a size cap (oldest removed first) because it runs on laptops; production retention follows §10.8 with case holds.

10.7 Secrets handling

Built: catalogue tokens and settings come from one local file (`prahari.env`) or environment variables, never from code; stream credentials are removed from URLs before logging or display; audit parameters named like `token`, `password`, `authorization` are redacted. Designed: secrets in a central vault or HSM, per-node certificates issued at enrolment and rotated automatically, revocation list checked at the gateway.

10.8 Data protection: DPDP Act 2023 and proportionality

Surveillance data about people's movements is personal data. Even where law-enforcement processing may fall within exemptions under the DPDP Act 2023, the design keeps the Act's principles, because the Supreme Court's *Puttaswamy* judgment requires any State intrusion into privacy to have a legal basis, a legitimate aim, proportionality and procedural safeguards.

Principle	Mechanism in PRAHARI	Status
Data minimisation	No central video; only plate events and small crops travel; FRS off by default	BUILT
Purpose limitation	Purpose and case ID mandatory for movement queries	BUILT
Accountability	Tamper-evident audit of queries, exports, watchlist and registry changes	BUILT
Accuracy	Alert vs review split; ranked evidence; implausible hops flagged; human acknowledgement	BUILT
Storage limitation	Tiered retention, enforced automatically (below)	DESIGNED
Security safeguards	§10.1–10.7	Partly built
Oversight	Auditor role; periodic audit reports; published impact assessment before FRS	DESIGNED

Proposed retention (for Department decision)

Data	Proposed default	Storage tier
Raw video	Not stored by PRAHARI; remains in the department recorder under its existing retention (7–15+ days)	Department
Plate events with no watchlist hit	90 days searchable, then compressed to 1 year, then deleted	Hot → warm
Evidence crops with no hit	90 days	Object store
Clips pulled for a case	30 days unless placed on case hold	Object store
Events, crops and clips linked to a case	For the life of the case and any statutory period (sizing assumes up to 7 years)	Cold, write-once
Audit log	Not deleted within the system's life	Replicated

10.9 Private cameras: consent tiers DESIGNED

Tier	What police get	Owner's controls
Registered (default)	Location and contact in the registry; footage requested manually for a case	Owner may leave at any time
Shared on request	Clip pull through a node for a specific case, with owner notification	Consent record with scope and expiry; owner can revoke; every pull audited and visible to the owner
Live-integrated	Camera analysed like a government camera (ANPR at a gate facing a public road)	Explicit consent artefact per camera; public-facing field of view only; revocable; periodic renewal

The registry already records every camera's source and ownership; the consent artefact, scope and revocation flag are fields added to the camera record in the pilot.

11 Deployment architecture

11.1 Physical topology

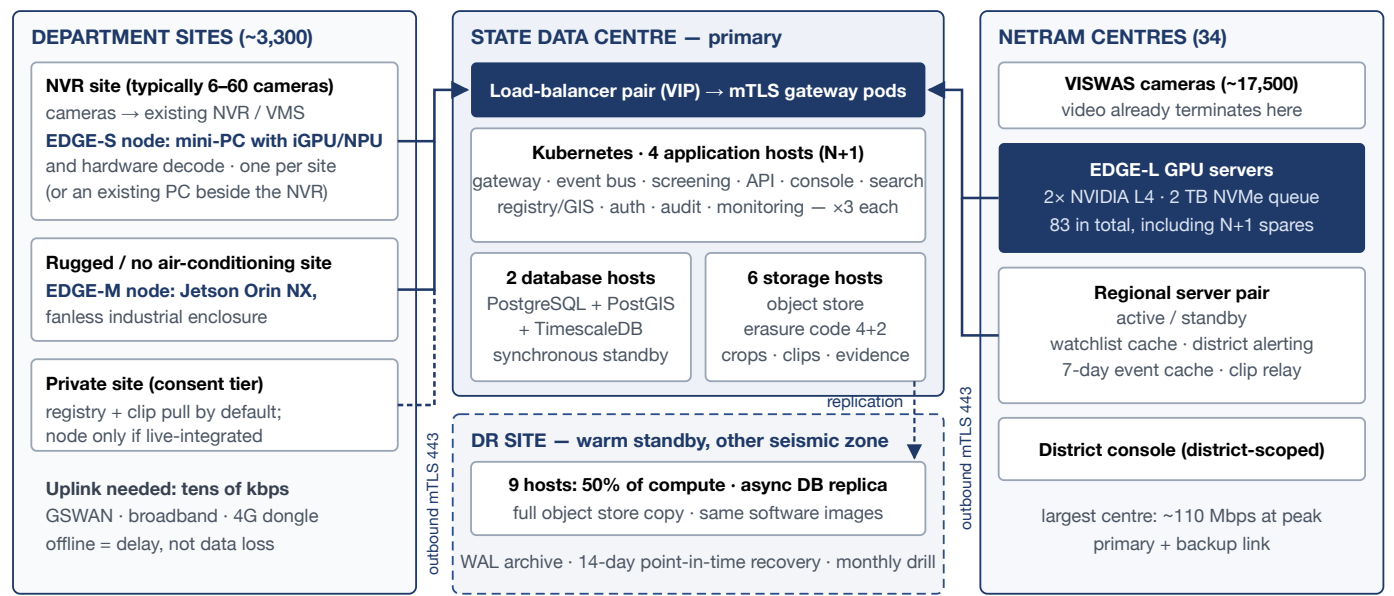


Figure 11.1 — Deployment topology for ~80,000 cameras. Host and node counts are from the sizing model (§12) **derived**; production node classes have not yet been benchmarked on the target hardware.

11.2 Deployment units

Unit	Where	Role	Count (§12)
EDGE-S node	Departmental NVR sites	Adapters, decode (Quick Sync), ANPR on iGPU/NPU, light analytics, disk queue. Runs on an existing PC beside the NVR where one is available.	~3,455
EDGE-M node	Rugged or unconditioned sites	Same software on a fanless Jetson Orin NX with hardware H.265 decode	as needed
EDGE-L server	Netram centres	Regional node for VISWAS and for sites without local compute; batched GPU inference; 2 TB queue	83 (166 L4)
Regional server pair	Netram centres	District alerting with a local watchlist cache (works through a central outage), 7-day event cache, clip/live relay, district console	68
Central application hosts	State Data Centre	Kubernetes: gateway, bus, screening, API/console/search, registry/GIS, auth, audit, monitoring	4
Central database hosts	State Data Centre	PostgreSQL + PostGIS + TimescaleDB, primary + synchronous standby (standby serves search/trace reads)	2
Object-storage hosts	State Data Centre	Crops, clips, case evidence; erasure coding 4+2; object lock for case evidence	6
DR site	Second Government DC or national government cloud, different seismic zone	Warm standby: 50% compute, async database replica, full object store	9 hosts

11.3 Software delivery and node lifecycle

Concern	Approach	Status
Install	Reference build: <code>pip install -r requirements.txt</code> , one settings file, <code>python run.py</code> ; <code>python run.py doctor</code> checks Python, packages, models, disk, database, catalogue and opens two real streams, with a fix for every failure. Runs on macOS, Windows and Linux, offline after install.	BUILT
Packaging	Signed OCI images for node and central services; an appliance image for EDGE-S/M; offline installation bundle for air-gapped sites.	DESIGNED
Node enrolment	Department admin issues a one-time enrolment token in the registry → node generates a key and certificate request → State CA issues a certificate bound to the site → node downloads its camera assignment from the registry and starts. No manual camera configuration on the node.	DESIGNED

Concern	Approach	Status
Updates	Staged rings (pilot district → one district per department → all), health-gated, with automatic rollback. Models versioned; the model version is recorded with each detection.	DESIGNED
Environments	Practice grid (built-in synthetic cameras, separate database) for training and demos; replica grid for soak and failure tests; staging with recorded fixtures of each VMS adapter; production.	Practice/replica BUILT
Field support	One field engineer per district, 5% cold spares; a failed node degrades analytics only — the NVR keeps recording.	DESIGNED

12 Infrastructure sizing and cost-benefit

This section is generated from the team’s sizing model and reproduced here with section numbers and cross-references adapted to this document. Scaling mechanics, high availability and disaster recovery from the same model are presented in §13.

The sizing model is generated from `bench/sizing_model.py`. Every table below comes from that script. To test a different assumption, edit the field in the `Assumptions` class and re-run:

```
.venv/bin/python bench/sizing_model.py # prints every table and writes docs/hld/sizing.json
```

Labels. Every number has one of four labels. `measured` means a file or benchmark in this repository, cited by path. `given` means it comes from the organisers or the public record. `price` means a market price, cited in §12.11. `assumed` means our estimate, with its basis and a confidence rating. Anything else is `derived`, and the formula is shown. All money is in ₹, **excluding 18% GST**. 1 lakh (L) = 10⁵ and 1 crore (Cr) = 10⁷. The figures are accurate to about ±30% and are rounded to match. We do not claim more precision than that.

12.1 Headline numbers (80,000 cameras, base case)

	PRAHARI (federated)	Model 4 (central VMS + AI)
Sustained backbone	0.21 Gbps avg / 0.45 Gbps peak	160 Gbps at 2 Mbps; 320 Gbps at 4 Mbps
Ratio	~360x at peak (750x on averages)	
Plate events	124 M/day. 1,400/s average, 3,100/s peak	same
Edge / GPU compute	83 GPU servers (166 L4) at 34 Netram + 3,455 edge mini-PCs	2,000 GPUs as usually specified (224 L4s for the same analytics) + 445 recording servers
Central storage	~210 TB logical / ~630 TB raw (all copies, erasure coding and DR included)	52 PB for 30 days of video, one copy
Capex	₹49 Cr	₹101 Cr (best case) to ₹175 Cr (as specified, infrastructure only). ₹312 Cr with commercial licences
Opex per year	₹18 Cr	₹84–95 Cr (infrastructure only). ₹119 Cr with licence support
5-year TCO	≈ ₹130 Cr	≈ ₹510–640 Cr (infrastructure only). ≈ ₹870 Cr with licences
Per camera	₹6,100 capex; ₹3,300 per year all-in over 5 years	₹12,700–21,800 capex; ₹12,800–16,000 per year

The fairest single comparison sets PRAHARI against the Model 4 best case: the same analytics, no licences, and only the GPUs it actually needs. That comparison is **₹130 Cr vs ₹510 Cr over 5 years, about 4x**. The difference is mostly the WAN bill, which recurs every year.

12.2 Reconciliation with the first-principles estimates

Earlier claim	Status	Corrected figure
First-principles estimate (§3.3): "160 Gbps → 0.39 Gbps, ~400x"	The arithmetic is correct for its inputs. Recomputed: 24,000 cams × 0.2 veh/s = 415 M events/day. At 200 B + 10 KB each, that is 392 Mbps, or 410x. But the inputs are off, and in both directions. The estimate assumed 3.4x too many plate events: 0.2 veh/s around the clock on every traffic-facing camera, including departmental gates. It also assumed 1.5x too small a crop. It left out telemetry, clip pulls and protocol overhead. And it compared <i>average</i> flows, while a link has to be provisioned for the peak.	Average ≈ 0.21 Gbps; peak hour ≈ 0.45 Gbps. The ratio is ~750x on averages and ~360x at peak. For a link you have to buy, peak is the honest basis. Across every sensitivity case the ratio stays between 110x and 5,500x. Recommended headline: "160 Gbps → under 0.5 Gbps at peak (~350x)". The conclusion stands. The number changes.
First-principles estimate (§3.2): "~2,000 central GPUs vs ~0"	Overstated. 2,000 GPUs buys continuous analytics at full video rate. A central system that ran the <i>same</i> 1-fps ANPR + 0.2-fps light analytics as PRAHARI would need about 224 L4s . In the central case, decode is the limit, not inference. PRAHARI also has GPUs: 166 L4s at the Netram centres , plus about 3,450 small edge nodes.	The case against Model 4 does not rest on GPUs. It rests on WAN bandwidth (≈ ₹60 Cr/yr) and 52 PB of video storage . Those two costs remain even in the most favourable version of Model 4 (§12.8).
README of the reference build: "roughly 40 cameras at one frame per second" per machine	That is the naive figure: 1000 / 21 ms = 47. It is correct for synthetic frames on the M1.	Planning figure: ~14 ANPR streams per M1-class worker at 1 fps , after a ×2 safety factor for real footage and 60% target utilisation (§12.5).

Earlier claim	Status	Corrected figure
README of the reference build: 48-camera stability figures (43% / 369 MB keyframes; 181% / 581 MB full decode)	Superseded: the 48-stream runs were repeated and saved in docs/hld/evidence/stability48-*.json (§12.7.3): keyframes ≈ 49% of a core, ~400 MB; full decode at 5 fps ≈ 168% of a core, ~560 MB; 48/48 live, 0 reconnects.	This model uses the 12-camera files. They agree per camera: full decode 3.7% vs 3.8% of a core per camera.

12.3 Assumptions

Only the assumptions that matter are listed here. The full list, with comments, is at the top of `bench/sizing_model.py`.

#	Assumption	Value	Label and basis	Confidence
A1	Cameras statewide	80,000	given : organisers	high
A2	VISWAS cameras (police)	17,500	given : ph-I ~7,000 at ~1,200 junctions/entry points in 41 places + ph-II ~10,500 (Surat, Vadodara, 52 municipalities, 80 border points)	high
A3	VISWAS video already terminates at the 34 Netram centres	yes	assumed : Netram is the district command-and-control centre that views VISWAS feeds	medium-high
A4	Traffic-facing (ANPR-viable) share, VISWAS / departmental	60% / 20% → 29% overall	assumed : VISWAS is sited at junctions and entry points. Departmental cameras are mostly corridors, counters and yards (§2.4, A6). CCAP measures this per camera (plate legibility, plate px height), so the registry discovers the true share in the first weeks of Phase 1. It is not a planning guess we are stuck with.	medium / low
A5	Vehicles per ANPR camera per day, VISWAS / departmental	12,000 / 2,000	assumed : one approach of an urban junction averages ~500/h over the day. Gates and frontage roads carry far less	medium / low
A6	Read yield (deduplicated plate events per vehicle pass)	0.8	assumed : two-wheeler plates are rear-only; occlusion; night	medium
A7	Peak-hour share of daily traffic (K-factor)	9% → peak/avg = 2.16	assumed : standard traffic-engineering range 8–10%	high
A8	ANPR analysis rate	1 fps at junctions; 4 fps at the 10% of cameras on highways and entry points → mean 1.3 fps	derived : at 60 km/h a vehicle crosses a ~15 m view in 0.9 s, so 1 fps can miss it	medium
A9	Light analytics (person / crowd / intrusion)	0.2 fps, 50 events/cam/day, 10% with an image	design / assumed	low
A10	Event size on the wire	550 B (+15% TLS/HTTP framing)	measured : mean JSON of stored plate events incl. 5 ranked candidates, data/demo.db, 281 rows = 547 B	high
A11	Evidence crop (plate + vehicle context)	15 KB	assumed , measured : synthetic plate crops average 5.2 KB (360×141, 2,196 files in data/evidence/). Real footage is noisier, and a vehicle crop is added	medium
A12	Clip pulls	2,000/day × 30 s × 4 Mbps	assumed : ~60 per district per day	low
A13	ANPR cost per frame (M1)	21 ms p90 at 960 px with a plate; 3.8 ms without; 40.8 ms at 1280 px	measured : prahari.anpr.bench, saved in docs/hld/bench-anpr-m1.json (CoreML detector, CPU OCR)	high (for M1, synthetic)
A14	Software decode cost (M1)	0.39% of one core per Mpix/s → 20% of a core per 1080p25 stream	measured : 12 cams, 111.7 Mpix/s, 43.8% CPU, data/stability-20260924-225251-full15-22642.json	high (for M1)
A15	Keyframe-only decode (M1)	0.52% of a core per camera	measured : data/stability-20260924-224549.json (12 cams, 6.2%)	high
A16	Production safety factor on per-frame cost	×2	assumed : real 1080p footage, more plates per frame, night, bigger production models	medium
A17	Target utilisation	60%	design: headroom for bursts, reconnect storms, rebuilds	high
A18	ANPR cameras whose GOP ≤ 1 s (keyframe decode is enough)	30%	assumed : The rest need full decode. CCAP reads the GOP from the stream	low
A19	Departmental NVR sites	50% × 6 cams, 35% × 20, 15% × 60 → mean 19 → ~3,300 sites	assumed : The registry replaces this with real data	low
A20	Relative performance vs one M1 worker	EDGE-S ×1.5, EDGE-M ×2, EDGE-L (2× L4) ×20	assumed : not measured on the target hardware	low-medium
A21	Hardware decode capacity (1080p25 streams)	EDGE-S 24, EDGE-M 18, EDGE-L 256	Orin NX: 18× 1080p30 H.265 (NVIDIA datasheet). L4: 4 NVDEC (datasheet) × ~32 streams each (assumed). QuickSync 24 (assumed)	medium
A22	Event row on disk (hot)	1,050 B incl. indexes	measured upper bound: data/demo.db 294,912 B / 281 rows	medium
A23	Retention	events hot 90 d, warm to 1 y; crops 90 d; clips 30 d; case-linked events, crops and clips cold for 7 y	policy (§10.8). We deliberately do not keep all plate reads for 7 years (DPDP proportionality). The UK NAS keeps them 1 year	policy

#	Assumption	Value	Label and basis	Confidence
A24	Model 4 video	2 Mbps/cam, 30 d retention, 40 streams/GPU, 600 Mbps per recorder, 4,000 WAN sites	assumed (first-principles estimate, \$3.2)	medium
A25	WAN price, Model 4	₹250 per Mbps-month on 1.25× sustained	price : ILL ₹25–38k/month for 100 Mbps (₹250–380/Mbps); ₹1.5 L for 1 Gbps (₹150/Mbps)	low: the biggest swing factor (\$12.9)
A26	DC facility	₹15,000 per kW IT per month, all-in	price : Indian colocation ₹8k–25k/kW-month. Cross-check: ₹9/kWh × 730 h × PUE 1.6 = ₹10.5k for power alone	medium
A27	Power	₹9/kWh	price : Gujarat HT effective ₹7.5–8.5/kWh; LT commercial is higher	medium
A28	AMC	8% of hardware per year, from year 2	assumed : typical Indian government AMC after warranty	medium
A29	Staff (loaded, per year)	12 NOC/SOC @ ₹15 L; 33 field engineers @ ₹8 L; PRAHARI platform team 15 @ ₹25 L; Model 4 +12 DC ops	assumed	medium
A30	Commercial licences (Model 4 only)	VMS ₹5k/channel, ANPR ₹30k/channel, basic analytics ₹5k/channel, +20%/yr support	assumed : low . Shown separately so no conclusion depends on it	low

12.4 Bandwidth

12.4.1 Statewide and per district

quantity	value	how
camera mix	23,000 ANPR + 57,000 light (29% ANPR)	A2 × A4
events/day	124 M plate + 2.8 M light	ANPR cams × A5 × A6
events/s	1,400 avg / 3,100 peak hour	÷ 86,400; × 2.16
GB/day	events 68 · crops 1,820 · telemetry 86 · clips 30 → ~2.3 TB/day	× A10, A11, 100 bps/cam, A12
events only (text, no images)	7 Mbps statewide average	
everything	210 Mbps avg / 450 Mbps peak	
mean district (1/33)	6.4 Mbps avg / 13 Mbps peak	
largest district (15% of cameras)	~67 Mbps peak	
largest Netram (Surat or Vadodara, ~30% of VISWAS)	~110 Mbps peak	
Model 4 video	160 Gbps at 2 Mbps; 320 Gbps at 4 Mbps; 1.73 PB/day	80,000 × bitrate
ratio	~750× average; ~360× peak	

Crops are 80% of PRAHARI's traffic. The bandwidth ladder therefore controls cost mainly through crops. At tier 2 (crops only for watchlist hits), statewide peak falls to **~30 Mbps**.

12.4.2 A departmental site, by bandwidth-ladder tier

site	share of sites	tier 3/4 avg	tier 3/4 peak	tier 1 (events only, zstd)	tier 0: offline days a 100 GB queue holds	a 30 s clip at 512 kbps
6 cams	50%	4 kbps	25 kbps	0.2 kbps	> 1 year	4 min
20 cams	35%	13 kbps	83 kbps	0.6 kbps	> 1 year	4 min
60 cams	15%	38 kbps	250 kbps	1.7 kbps	240 days	4 min

What this means for the low-bandwidth strategy. A departmental site needs **tens of kbps, not Mbps**. A 4G dongle or an existing GSWAN or broadband drop is enough. For a departmental site, "offline" is a latency problem. It is not a data-loss problem. The queue does not fill. Only the Netram nodes carry real volume. The largest Netram averages ~50 Mbps, or ~550 GB/day (**derived**: 110 Mbps ÷ 2.16 × 86,400 s ÷ 8). **EDGE-L therefore gets a 2 TB NVMe queue (~3.5 days of outage)**. Clip pulls are the only transfer that needs bandwidth. On a thin link the clip arrives in minutes rather than seconds, and the console shows its progress.

Tier	Uplink	Sent	Statewide peak at this tier
4	> 10 Mbps	events + crops + thumbnails + live relay on request	~450 Mbps + relays
3	2–10 Mbps	events + crops	~450 Mbps
2	0.2–2 Mbps	events; crops only for watchlist hits	~30 Mbps
1	< 200 kbps	events only, zstd-batched	~6 Mbps
0	offline	disk queue; replays in order on reconnect	0

12.5 Edge compute

12.5.1 Streams per node, derived from our measured per-frame time

Formula (node_streams): **streams = 1000 ms × util × perf ÷ (ms per frame × safety × fps)**. The number of full-decode ANPR streams is also capped at hw_decode × util ÷ (1 – keyframe share).

class	hardware	price (price /ASSUMED)	power	ANPR @1 fps	ANPR @ mixed 1.3 fps	light-only @0.2 fps
M1 (measured)	1 worker	–	–	47 naive → 14 planned	–	–
EDGE-S	Mini-PC, Intel Core Ultra (iGPU + NPU + QuickSync), 16 GB, 512 GB SSD	₹75k (₹55k–1 L)	35 W	20	16 (inference-bound)	~590
EDGE-M	Jetson Orin NX 16 GB, fanless, 512 GB NVMe (rugged, no-AC sites)	₹1.1 L (module ₹49k)	25 W	15	15 (decode-bound)	~790
EDGE-L	2U server, 2× NVIDIA L4, 256 GB, 2 TB NVMe queue (at Netram)	₹11.5 L	650 W	219	219 (decode-bound)	~7,900

Decode, not inference, is the edge bottleneck. In software on the M1, one 1080p25 stream costs 20% of a core (measured, A14). Eight cores therefore full-decode only ~23 streams at 60% utilisation. Keyframe-only decode costs ~40× less, but it gives 1 fps only when the camera's GOP is ≤ 1 s. So every production node class has a hardware decoder, and CCAP records each camera's GOP. Where a department allows it, a one-time camera setting change to GOP = fps would double EDGE-L capacity. That is sensitivity case "All ANPR cams keyframe-decodable". Light analytics is almost free: a single node can carry hundreds of those cameras.

12.5.2 Fleet for 80,000 cameras

item	count	how
EDGE-L at Netram (VISWAS)	49 working + 34 N+1 spares = 83 servers (166 L4)	max(inference, decode) on 10,500 ANPR + 7,000 light, + 1 per Netram
Departmental NVR sites	~3,300 (mean 19 cams)	A19
EDGE-S	3,290 + 5% spares = 3,455	one per site. The count is set by the number of sites, not by load. Even a 60-camera site (12 ANPR) fits on one node
Edge power	~170 kW, spread across 3,300 premises	

The departmental edge costs ~₹26 Cr in hardware. We have assumed that every site buys a new node. Where a department already has a PC beside its NVR (i5-class or newer), PRAHARI runs on that PC. At 40% reuse, capex falls by ₹12 Cr (§12.9).

12.6 Regional (Netram) and central (SCRB / State Data Centre) tiers

12.6.1 Workload

	value	how
Ingest	1,400 events/s avg, 3,100/s peak	\$12.4
Rows per year	~ 45 billion (124 M/day × 365)	
Hot rows (90 d)	~11 billion, ~ 12 TB per copy	× 1,050 B (A22)
Warm rows (to 1 y, compressed ×5)	~7 TB per copy	
Watchlist screening	3,100 events/s × 0.5 ms confusion-aware match ≈ 1.6 cores (3 pods, for HA)	assumed 0.5 ms/event
Search / trace	10,000 queries/day, ~0.5/s peak. Each fuzzy query expands to ≤ ~64 candidate plates and probes a (plate, ts) index in daily partitions	assumed
Monitoring	~1.5 M time series (3,538 nodes × 200 + 80,000 cams × 10)	assumed per-node/per-cam series

At this ingest rate a single PostgreSQL + TimescaleDB primary is enough. It needs daily chunks, a (plate, ts) btree, and a trigram index for partial plates. The write rate is under 10% of what one tuned node handles with batched inserts. Sharding (Citius, keyed by plate hash) becomes necessary only at roughly 10× today's event rate. A columnar store such as ClickHouse is a reasonable alternative for the warm tier. The model does not depend on that choice.

12.6.2 Hardware

tier	contents	count
Central DC (primary)	Kubernetes: ingest gateway ×3, event bus (NATS JetStream / Kafka) ×3, screen/match ×3, API + console + search ×3, registry/GIS/auth/audit ×3, monitoring ×4, control plane ×3 = 220 vCPU	4 app hosts (2-socket, 512 GB, N+1)
	PostgreSQL + PostGIS + TimescaleDB: primary + synchronous standby (the standby serves search/trace reads)	2 DB hosts , ~19 TB NVMe each
	Object store (MinIO / Ceph, erasure coding 4+2) for crops, clips and case evidence	6 storage hosts (12 × 20 TB)
DR site (warm standby)	50% of app compute, 1 async DB replica, full object store	9 hosts
Regional: 34 Netram	2 servers per Netram (active/standby): district relay, local watchlist cache and alerting , district console, 7-day event cache, clip relay. Plus the EDGE-L servers from §12.5	68 servers + 83 EDGE-L
Central IT power (DC + DR)	~12 kW	

The central tier is small because PRAHARI moves no video to it. That is deliberate. The whole central footprint for statewide PRAHARI (DC + DR, 21 hosts) is smaller than **one row** of the Model 4 design, which needs 500 GPU servers and 445 recorders.

12.7 Storage tiers

tier	TB (one logical copy)	medium	copies
events hot, 90 d	12	NVMe, row store	3 (primary, standby, DR)
events warm, 90 d – 1 y	7	NVMe/SSD, compressed chunks	3
events cold, case-linked, 7 y	0.07	object store (HDD)	EC 4+2 × 2 sites
crops hot, 90 d	163	object store	EC 4+2 × 2 sites
crops cold, case-linked, 7 y	5	object store	EC 4+2 × 2 sites
clips hot, 30 d	1	object store	EC 4+2 × 2 sites
clips cold, case-linked, 7 y	23	object store	EC 4+2 × 2 sites
event bus, 7-day replay	1.4 (incl. ×3)	NVMe	×3
total	~210 TB logical / ~630 TB raw		
<i>departmental video</i>	not stored centrally. It stays on departmental NVRs at their own 7–15-day retention (given), FAQ). Clips are pulled by case ID		
<i>Model 4 video, 30 d</i>	51,800 TB = 52 PB (one copy; ~3,240 × 20 TB drives before overhead)		

Crops account for 77% of PRAHARI's central storage. Keeping every plate read for 7 years instead of 1 would add ~15 PB raw. It would also be disproportionate under the DPDP Act. So only case-linked records go to the cold tier.

12.8 Costs

12.8.1 Capex, ₹ crore, 80,000 cameras (ex-GST)

PRAHARI	₹ Cr	Model 4 (as specified, infrastructure only)	₹ Cr
edge nodes + installation (83 EDGE-L, 3,455 EDGE-S, ₹15k install per node)	40.5	500 GPU servers (2,000 L4)	82.5
central DC (hosts, NVMe, object store, network)	2.5	445 recording servers	28.9
DR site	2.2	52 PB video storage	37.3
regional (68 Netram servers)	3.4	core network (100G spine)	8.0
		site WAN installs (4,000 × ₹30k)	12.0
		metadata / DR / management	6.0
PRAHARI capex	48.6	Model 4 capex	174.7
		+ commercial VMS/ANPR/analytics licences, if bought	+137.5

12.8.2 Opex per year, ₹ crore

PRAHARI	₹ Cr/yr	Model 4 (infrastructure only)	₹ Cr/yr
edge power (170 kW across sites)	1.3	WAN bandwidth for video (160 Gbps × 1.25 × ₹250)	60.0
regional power	0.3	DC facility (~630 kW IT)	11.4
DC + DR facility (~12 kW IT)	0.2	hardware AMC	14.0
central uplinks (2 × 1 Gbps at DC and at DR)	0.7	staff (NOC/SOC, DC ops, field)	6.2
Netram backup links (34 × 100 Mbps)	1.2	in-house platform team	3.8
new links for the 30% of sites without one	1.8		
hardware AMC	3.9		
staff (platform 15, NOC/SOC 12, field 33)	8.2		
total	17.6	total	95.4 (+ licence support 27.5 if licensed)

Staff is PRAHARI's largest opex line, at 47%. Its infrastructure costs are small. Model 4's largest line is the WAN, at 63%. **A field fleet of 3,455 edge nodes is PRAHARI's real operational burden.** At a 3–5% annual failure rate that means ~100–170 site visits per year. We budget one field engineer per district plus 5% cold spares, and the NVR keeps recording while a node is down.

12.8.3 Five-year TCO (capex + 5 years opex; AMC from year 2), ₹ crore

design	capex	opex/yr	5-yr TCO
PRAHARI (federated)	48.6	17.6	~133
Model 4 best case : same 1-fps analytics, 224 L4s, infrastructure only	101	83.5	~511
Model 4 as usually specified: 2,000 GPUs, infrastructure only	175	95.4	~638
Model 4 as usually specified + commercial licences	312	119	~866

12.8.4 Phased rollout

phase	months	cameras	EDGE-L	EDGE-S	plate events/day	peak Mbps	capex this phase	opex/yr at end of phase
1: VISWAS + 34 Netram	0–6	17,500	83	0	~100 M	~370	₹17.7 Cr	₹12.5 Cr
2: + departments	6–18	50,000	83	~1,800	~110 M	~410	₹16.1 Cr	₹15.1 Cr
3: 80k + private opt-in	18–36	80,000 + 10,000 private (registry + clip pull only)	83	~3,450	~124 M	~450	₹14.8 Cr	₹17.6 Cr

Phase 1 carries the whole central core, DR and all 34 Netram deployments. Staff costs are full-size from day 1. **VISWAS produces ~80% of all plate events.** The departments add many cameras but few plates. So traffic and central sizing are essentially settled in Phase 1, and Phases 2–3 are mostly a field-deployment exercise.

12.9 Sensitivity (80,000 cameras; TCO = 5 years, ₹ crore)

case	plate events/day	PRAHARI peak Mbps	ratio @ peak	edge nodes	central raw TB	PRAHARI capex	PRAHARI TCO	M4 best case TCO	M4 infra TCO	M4 + licences TCO
Base case	124 M	450	360×	3,538	630	49	133	511	638	866
ANPR-capable share 50% (not 29%)	180 M	650	250×	4,072	900	55	143	522	638	947
ANPR at 2 fps everywhere	124 M	450	360×	4,089	630	57	145	521	638	866
Crop 10 KB	124 M	310	520×	3,538	470	49	133	511	638	866
Crop 30 KB	124 M	860	190×	3,538	1,100	49	133	511	638	866
Traffic ×2	240 M	880	180×	3,538	1,200	49	133	511	638	866
First-principles traffic (0.2 veh/s on every ANPR cam)	400 M	1,400	110×	3,538	1,900	49	134	511	638	866
Safety factor 3 (real footage worse)	124 M	450	360×	4,080	630	56	144	519	638	866
All ANPR cams keyframe-decodable (GOP ≤ 1 s)	124 M	450	360×	3,538	630	49	133	511	638	866
Reuse existing departmental PCs at 40% of sites	124 M	450	360×	2,222	630	37	117	511	638	866
Crops only for watchlist hits (tier 2)	124 M	29	5,500×	3,538	130	48	132	511	638	866
Cold 7 y for all events (not recommended)	124 M	450	360×	3,538	15,000	57	147	511	638	866
WAN ₹100/Mbps-month (state-owned fibre)	124 M	450	360×	3,538	630	49	133	331	458	686
WAN ₹500/Mbps-month (small site links)	124 M	450	360×	3,538	630	49	133	811	938	1,166
Model 4 at 4 Mbps	124 M	450	720×	3,538	630	49	133	920	1,046	1,275
Model 4 at 7-day retention	124 M	450	360×	3,538	630	49	133	471	597	826

How to read this table:

- PRAHARI's TCO barely moves (₹117–147 Cr) across every case.** Its cost comes from the number of sites and from staff, not from traffic. Traffic, crop size and ANPR share move *bandwidth* and *storage*, which are cheap at this scale.
- Model 4's TCO is dominated by the price per Mbps.** If the state can deliver 160 Gbps over its own fibre (GSWAN / BharatNet) at ~₹100/Mbps-month, the best case falls to ~₹330 Cr, **still ~2.5× PRAHARI**. That is the most favourable case for Model 4 in this model. Even then it needs 52 PB, and it needs every department to send its video out, which is the consent problem in §3.2.
- The bandwidth *ratio* ranges from 110× (first-principles traffic) to 5,500× (tier 2). It never approaches parity.

The five assumptions that move the result most:

- WAN price per Mbps (A25).** It swings Model 4's 5-year cost by ±₹300 Cr.
- Number of departmental NVR sites (A19).** It sets PRAHARI's node count, edge capex and field workload. The registry replaces this assumption with real data.
- Real-footage performance vs the M1 (A16, A20).** None of our production node classes has been measured. A safety factor of 3 adds ~550 nodes and ~₹11 Cr to the 5-year TCO.
- VISWAS vehicles per camera and ANPR share (A4, A5).** Together they drive ~80% of events, bandwidth and crop storage.
- Crop size and crop policy (A11, ladder tier).** Crops are 80% of the bytes. Sending them only for hits cuts peak traffic 15× and central storage 5×.

12.10 What we measured vs what we assumed

Measured on an Apple M1, 8 GB, with **synthetic streams** and **synthetic plates**:

- ANPR cost per frame: 3.8 ms detector-only and 21 ms with one plate (p90, 960 px); 40.8 ms at 1280 px. Source: `docs/hld/bench-anpr-m1.json`.
- Software decode: 20% of a core per 1080p25 stream (scaled from 12 cameras at 111.7 Mpix/s = 43.8% CPU). Keyframe-only: 0.52% of a core per camera. Zero decode errors over 100–200 s runs. Source: `data/stability-*.json`.
- Event size: 547 B serialised. Evidence crop: 5.2 KB (plate only, synthetic). Stored row: ≤ 1.05 KB including indexes. Source: `data/demo.db`, `data/evidence/`.
- Practice mode: 8 cameras, 788 frames at ~ 14 ms average (`docs/hld/evidence/practice-run-runtime-2026-09-24.json`).

Not measured, and assumed with an explicit margin:

- Performance on EDGE-S, EDGE-M and EDGE-L hardware. None of it has been benchmarked.
- Real CCTV footage (night, glare, two-line plates).
- Real camera GOP settings, the number of cameras per departmental site, and actual traffic per camera.
- Every price. These are Indian retail and list anchors; government volume procurement is usually 15–30% lower, and we have **not** applied that discount.

The $\times 2$ safety factor and 60% utilisation are there because the first set is unmeasured. The model is set up so that each of the second set can be replaced by a registry measurement during Phase 1.

What would change our minds:

- Suppose a Phase-1 pilot measures production nodes at half our assumed inference *and* decode performance. EDGE-L grows from 83 to 132, EDGE-S from 3,455 to 3,973, and edge capex from ₹40 Cr to ₹51 Cr. The 5-year TCO rises from ₹133 Cr to ₹148 Cr (checked in the model). The departmental fleet grows very little, because its size is set by the number of sites, not by load.
- If the state already owns ≥ 160 Gbps of spare fibre capacity to a single DC, the WAN argument weakens to $\sim 2.5\times$. The storage and departmental-consent arguments would remain.

12.11 Sources (price and fact anchors, accessed 25 Sep 2026)

- NVIDIA L4 price in India: IndiaMART listing ₹1 L; IndiaMART listing ₹3 L; AceCloud, L4 price in India (₹2.4 L buy, ₹36.7k/month rent)
- NVIDIA L4 has 4 NVDEC and 2 NVENC decoders/encoders and a 72 W TDP: Cisco-hosted NVIDIA L4 datasheet
- Jetson Orin NX 16 GB module ₹49k: IndiaMART. Decode 18 $\times$ 1080p30 H.265: NVIDIA Jetson Orin NX series datasheet
- Server: Dell PowerEdge R760 (Xeon Gold 4409Y, 2 $\times$ 32 GB) ₹6.51 L: IndiaMART
- HDD: Seagate Exos X20 20 TB ₹89.9k–92k: Computech Store, IndiaMART
- Leased-line pricing in India (100 Mbps ₹25–38k/month; 1 Gbps ₹1.5 L+/month; ex-GST): itforsme.in ILL pricing guide
- Colocation in India (full rack ₹25k–60k/month; power ₹8k–25k per kW-month): Cyfuture colocation pricing
- Gujarat tariffs held at previous-year levels for FY 2026-27: Mercom India. HT energy charge $\sim$ ₹4.3/kWh, effective ₹7.5–8.5/kWh with FPPPA, duty and demand charges: Bridgeway Power, GERC HT tariff 2026-27
- VISWAS phase I ($\sim 7,000$ cameras, $\sim 1,200$ locations, 34 Netram): DeshGujarat, Jul 2022. Phase II ($\sim 10,500$ cameras, Surat/Vadodara, 52 municipalities, 80 entry/exit points): DeshGujarat, Dec 2023
- Departmental retention of 7–15 days and mixed cloud/local storage: organisers' FAQ (`research/00-official-faqs.txt`)
- **No source found; clearly labelled assumed ranges:** Intel Core Ultra mini-PC price (₹55k–1 L), per-channel VMS/ANPR licence prices, AMC rate, staff costs, and per-NVDEC stream throughput on the L4.

13 Scalability, availability and disaster recovery

13.1 Horizontal scaling units and load balancing

- **Edge.** One EDGE-S per departmental recorder site (up to ~ 16 ANPR cameras at mixed frame rates plus hundreds of light-analytics cameras); one EDGE-L per ~ 220 ANPR streams at a Netram centre, plus N+1. A new department is added as *more nodes*; nothing central changes. derived §12
- **Central.** Stateless pods (gateway, screening, API) scale on consumer lag, about one gateway pod per 2,500 events/s. The event bus is partitioned by `camera_id`, which preserves per-camera ordering and timestamp monotonicity. Events are partitioned by day in TimescaleDB; a plate-hash shard (e.g. Citus) is needed only at $\sim 10\times$ today's projected rate. The object store grows by one storage host per ~ 160 TB usable.
- **Load.** Base case 1,400 events/s average and 3,100/s peak statewide (124 million plate events a day) derived — well within the $\sim 12,000$ events/s ingest rate reported for the UK National ANPR Service.
- **Load balancing.** Nodes only dial out. Each holds an ordered endpoint list — State DC gateway VIP, then its Netram relay, then the DR gateway VIP. A pair of L4 load balancers (keepalived/HAProxy, or the State DC's own) fronts the gateway pods. Reconnection uses the same jittered exponential backoff as camera reconnection (2 s $\rightarrow$ 30 s), so a gateway restart cannot cause a reconnect storm. Camera-to-node assignment lives in the registry, so cameras of a failed node at a multi-node site move to a sibling.

13.2 Monitoring, logging and health checks

Item	Detail	Status
Per-camera health	State, measured fps, codec, resolution, frames decoded/emitted, reconnects, discontinuities, decoder warnings by kind, decode errors, last error, last-frame age, next retry; written to the registry every 10 s	BUILT
Service health	/api/health (database size and counts, disk, live subscribers); /api/runtime (cameras, plate-reader state and throughput, disk); run.py doctor	BUILT
Soak / stability harness	python -m prahari.node.stability: runs every catalogue camera, prints a table every 10 s, writes a JSON summary, exits non-zero if any camera never went live	BUILT
Metrics and logs at scale	Node heartbeat every 30 s; metrics by remote-write to VictoriaMetrics/Mimir (~1.5 million series), logs to Loki, Grafana dashboards alongside the registry map	DESIGNED
Alert rules	Camera offline > 5 min · node silent > 2 min · queue depth > 1 h · ingest lag > 60 s · clock skew > 2 s (timestamps are evidence)	DESIGNED

13.3 High availability

Component	Mechanism
Gateway, bus, screening, API	Three replicas across hosts, with quorum for the bus
PostgreSQL	Synchronous standby with automated failover (Patroni, < 60 s)
Object store	Erasure coding 4+2 across six hosts — survives two host failures
Netram tier	Active/standby regional pair; N+1 EDGE-L servers
District alerting	Continues through a central outage: each Netram holds a watchlist cache and screens its own district's events
Edge node	Single node per small site; if it fails, only analytics pause — the department recorder keeps recording, so no video is lost, and events resume with a new epoch

13.4 Disaster recovery

The DR site is a warm standby (full storage, 50% compute) in a different seismic zone from the primary — Kutch is in seismic zone V, so neither site should be there. Options are a second Gujarat Government data centre or the national government cloud.

Data / service	RPO	RTO	Why achievable
Plate events	effectively 0	1 h	Nodes keep events in their disk queue until the central store acknowledges them; the DR site receives the replay
Watchlist alerts within a district	0	~0	Netram screens locally; no central dependency
Cross-district alerts, search, trace	< 1 min	4 h	Asynchronous database replication; warm standby; DNS/VIP switch
Evidence crops	< 15 min	4 h	Bucket replication; edge ring buffer keeps 72 h as a second source
Case clips and evidence bundles	0 once sealed	4 h	Hash-sealed at the edge and replicated on seal; the department recorder holds the original
Audit log	0	4 h	Hash chain replicated; daily head-hash anchor exported offline

RPO/RTO values are design targets from the sizing model, to be proven in failover drills during Phase 1; they are not measured.

Backup. Continuous WAL archiving to the object store with 14-day point-in-time recovery; nightly logical dumps of registry and watchlists; object lock (write-once) on case-evidence buckets; monthly DR failover drill. **Cybersecurity controls** for all tiers are in §10.

14 Department-wise information requirements and prerequisites

The Problem Statement asks what information departments must provide so that integration feasibility can be assessed. PRAHARI's answer is to ask for as little as possible and **measure the rest**: once a camera's stream can be opened, the platform itself establishes its codec, resolution, real frame rate, health and whether it can read plates, and writes that to the registry.

14.1 What each department provides

#	Item	Why it is needed	Form	Mandatory
1	Nodal officer and technical contact	Approvals, access, escalation; owner of the department's node(s)	Name, designation, phone, e-mail	Yes

#	Item	Why it is needed	Form	Mandatory
2	List of sites with recorders (NVR/DVR/VMS) and approximate camera counts	Node placement and count; the largest driver of edge cost (§12)	CSV (site, address, recorder, cameras)	Yes
3	Network reachability at each site	A host on the camera network with outbound HTTPS to one State address; or a route to the nearest Netram centre	Yes/no per site; uplink type	Yes
4	Stream access	Read-only user on the recorder or cameras; RTSP URL pattern or VMS API access; sub-stream if available	Credentials via secure channel (never e-mail)	Yes
5	Camera location	Map, coverage-gap analysis and trace plausibility	Lat/long, or address to be geocoded and verified	Yes
6	Host for the node	An existing PC/VM beside the recorder, or space and power for an edge box	Yes/no; spec if existing	Yes
7	Legal basis and consent (private cameras)	Consent tier and scope (§10.9)	Consent artefact	Private only
8	Recorder retention and export method	Clip-on-demand within the retention window	Days; VMS product/version	Recommended
9	Time synchronisation	Timestamps are evidence; NTP source on the recorder/camera network	NTP server or “none”	Recommended
10	AMC vendor contact	Faults found by health monitoring can be raised directly	Vendor, contract end date	Optional

14.2 What PRAHARI measures instead of asking

Property	How it is obtained	Status
Codec, resolution, real frame rate	From the stream itself (decoder, PTS)	BUILT
Online / degraded / offline, uptime, reconnect history, last error	Capture worker statistics every 10 s	BUILT
Decoder health (warnings, errors)	Counted per camera by kind	BUILT
Plate legibility and ANPR viability, with reason	Capability profiler (§8.5)	BUILT
Coverage gaps, cameras without location or stream, ageing/stale cameras	Gap report over the registry	BUILT
GOP length, bitrate, clock skew, scene type	Stream analysis; ONVIF device time; scene classifier	DESIGNED

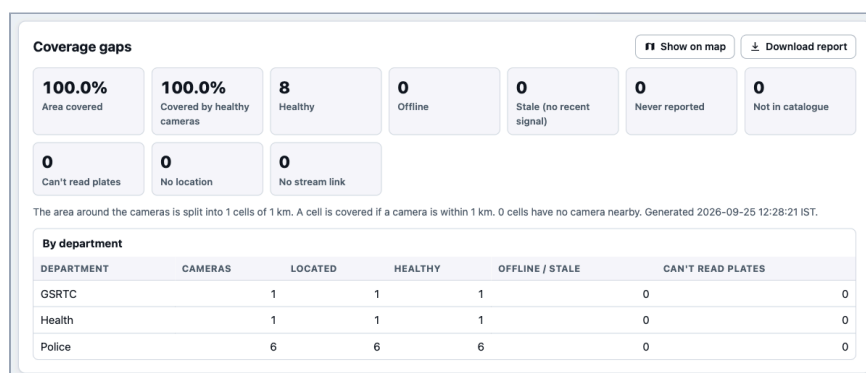


Figure 14.1 — Coverage-gap report (practice mode): uncovered cells, camera health and ANPR-unviable cameras by department, produced from the registry and measured capability.

14.3 Department profiles

Department (examples)	Typical camera settings	Primary use in PRAHARI	Specific prerequisite
Home / Police (VISWAS)	Junctions, city entry/exit points, pilgrimage centres	ANPR, watchlist alerts, trace; the bulk of plate events	Rack space and power at each Netram centre for EDGE-L and the regional pair
Transport / RTO	Offices, test tracks, check-posts	ANPR at check-posts; blacklist alerts	Check-post uplinks (often thin: ladder tiers 1–2)
Food & Civil Supplies	Godowns, PDS shops	Vehicle in/out ANPR at gates; after-hours intrusion (roadmap)	Many small sites: EDGE-S or an existing PC
Health	Hospital corridors, OPD counters	Health monitoring; crowd density (roadmap); rarely ANPR	Patient-privacy review before any analytic beyond health monitoring

Department (examples)	Typical camera settings	Primary use in PRAHARI	Specific prerequisite
GSRTC	Bus stations, depots, platforms	ANPR at depot gates; crowd and person analytics (roadmap)	Depot recorders usually centralised per division
Panchayat / Rural Development	Offices, gates, village squares	Health monitoring; ANPR where a gate faces a road	Weakest uplinks: store-and-forward essential
Municipal corporations	City surveillance, traffic, public spaces	ANPR, trace; crowd (roadmap)	Existing city command centres can host regional nodes
Private (societies, malls)	Gates, parking, frontage	Registered by default; clip on request; live only by consent	Consent artefact; public-facing view only

15 Statewide rollout plan and future roadmap

15.1 Phases

The phasing follows the sizing model (§12). VISWAS produces about 80% of all plate events, so traffic and central sizing are settled in Phase 1; Phases 2 and 3 are mainly field deployment. Month 0 is the start of the engagement; timelines are indicative and depend on procurement and departmental provisioning.

Phase	Months	Scope	Exit criteria
0 — Pilot and validation (within Phase 1)	0–2	One Netram centre and its VISWAS cameras; two departments' sites in the same district; central core on a small footprint. Close the “designed” items needed for production: SSO/RBAC, node mTLS and store-and-forward, evidence hashing and manifests. Measure real-footage ANPR accuracy, node hardware throughput, the confusion matrix and matcher performance at 10 ⁵ watchlist entries.	Measured accuracy and latency reported per camera type; zero-loss replay through a forced outage; security review passed; field figures replace the synthetic ones in this document
1 — VISWAS + 34 Netram	0–6	Capex ₹17.7 Cr. ~17,500 VISWAS cameras through 83 EDGE-L servers at all 34 Netram centres; central DC and DR; registry opened to all 26 departments for inventory (Model 1 everywhere, before any streaming); eGujCop watchlist adapter (subject to provisioning)	All VISWAS cameras in the registry with measured capability; district alerting live; DR failover drill passed
2 — Departments	6–18	Capex ₹16.1 Cr. Departmental estates to ~50,000 cameras; ~1,800 EDGE-S nodes; ONVIF and first VMS adapters; bandwidth ladder at thin sites; VAHAN enrichment	Each department onboarded with its own health and gap report; per-site uplink within ladder budgets
3 — Statewide + private	18–36	Capex ₹14.8 Cr. ~80,000 cameras; ~3,450 EDGE-S nodes; ~10,000 private cameras on consent tiers (registry + clip pull); scene-classified non-ANPR analytics	Target availability met for a full quarter; independent audit of access logs and retention

15.2 Technical roadmap

#	Item	Current status	Phase
1	Retrain/fine-tune plate detector and OCR on real Indian CCTV, including two-wheeler and two-line plates; re-fit confusion costs from measured errors	Synthetic-only	0
2	Operator SSO, RBAC and department/district scoping; restricted CORS	Self-asserted actor	0
3	Node mTLS gateway link, disk store-and-forward, bandwidth ladder	DESIGNED	0
4	SHA-256 evidence hashing at the edge, signed manifests, offline-verifiable export, audit head anchoring	Audit chain built	0
5	Watchlist matcher index and benchmark at 10 ⁴ –10 ⁵ entries	Linear scan	0
6	PostgreSQL/PostGIS/TimescaleDB store, event bus, containerised central services, observability stack	SQLite, in-process	1
7	Government-database adapters (eGujCop, VAHAN, SARTHI) with record/replay tests	CSV/API import built	1–2
8	ONVIF discovery and VMS adapters; clip pull and on-demand live view	RTSP/HLS built	2
9	Re-ID bridging for unreadable plates; road-network travel times; vehicle make/colour cross-check	Plate-anchored trace built	2
10	Scene classification; person, crowd, intrusion, loitering analytics	ANPR viability built	2–3
11	FRS under the proportionate design of §8.7, only after a published impact assessment	Not built	3, if approved

16 Working system and test evidence

16.1 How to reproduce every result in this document

Command (from the repository root)	What it does
<code>python run.py demo</code>	Practice mode: starts a built-in 8-camera RTSP grid and catalogue, the full platform and console at <code>http://127.0.0.1:8000</code> , with a separate database
<code>python run.py doctor · python run.py</code>	Checks the machine and the real catalogue link; runs the platform on the catalogue in <code>prahari.env</code>
<code>python -m pytest tests/ -q</code>	Automated test suite (148 tests)
<code>python -m prahari.node.stability --catalogue URL --seconds N</code>	Ingest soak test; per-camera table and JSON summary
<code>python -m prahari.anpr.bench · python -m prahari.anpr.evaluate</code>	ANPR throughput and synthetic accuracy (§8.3–8.4)
<code>python bench/bench_plate_matching.py</code>	Exact vs confusion-space matching benchmark (§7.6)
<code>python bench/sizing_model.py</code>	Regenerates every sizing and cost table (§12)
<code>sandbox/rtsp_server.py</code> (+ <code>--loop-pts-reset</code> , <code>--join-mid-gop</code>)	Replica of the Sentinel grid's behaviours for failure testing; kill and restart it to simulate a gateway outage

16.2 End-to-end practice run

The practice grid (`sandbox/demo_grid.py`) has eight cameras in Gandhinagar: six along Ch-Road about 150 m apart, a hospital OPD corridor and a GSRTC depot, alternating H.264 and H.265, 960×540 at 10 fps, looping every 40 s. A stolen white Swift, GJ01AB1234, is scripted past cameras 1 → 2 → 3 → 4 about nine seconds apart; a second listed vehicle, GJ18BD6612, passes cameras 2 and 4; other traffic is random. The video is synthetic and exists to prove the plumbing end to end, not accuracy.

Check	Result measured , practice mode
Cameras onboarded from the catalogue and live	8 of 8 (including the corridor and depot cameras, which the profiler does not treat as plate cameras)
Trace of GJ01AB1234	Exactly its 4 scripted cameras, in order 1 → 2 → 3 → 4 , with timestamps and plausible hop speeds
Watchlist alerts	6 correct alerts (4 for GJ01AB1234, 2 for GJ18BD6612) and 0 false alerts
Plate reads and events	198 plate reads de-duplicated to 55 events; ~14 ms average per analysed frame (788 frames)

Runtime counters saved in `docs/hld/evidence/practice-run-runtime-2026-09-24.json` (59 s after start: 8/8 live, 0 reconnects, 788 frames analysed at 14.4 ms average, 198 plate reads → 55 de-duplicated events → 6 alerts, 0 review items). Repeat live with `python run.py demo`; it takes about a minute to show the same result. Because clips loop, a later trace shows repeat visits (for example stops 1, 5, 9 at the same camera), which is expected.

16.3 Screens of the working platform

All screenshots are of the running reference implementation in practice mode (light theme). Trace (Figures 8.1–8.2), Alerts (Figure 9.2), the coverage-gap report (Figure 14.1) and the API documentation (Figure A.1) appear in their sections.

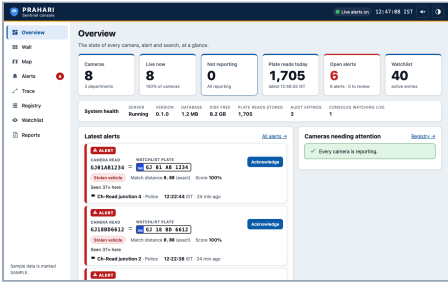


Figure 16.1 — Overview: cameras live/offline, plate reads, open alerts, plate-reader health.

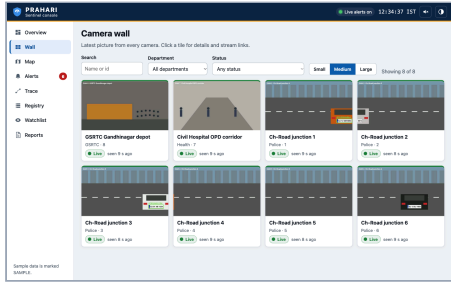


Figure 16.2 — Wall: latest analysed frame per camera with plate boxes; filters by department and state.

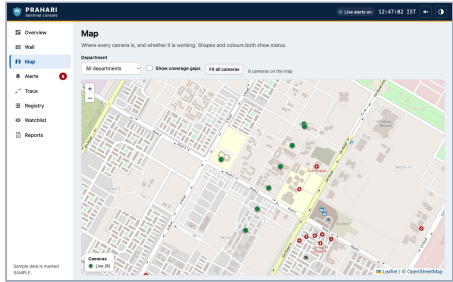


Figure 16.3 — Map: every camera by health, department filter, coverage-gap layer.

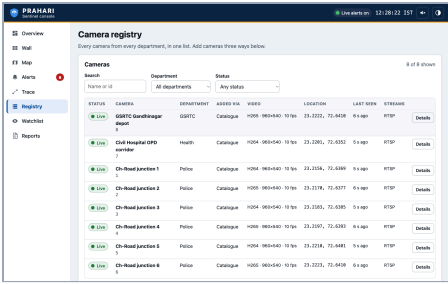


Figure 16.4 — Registry: all cameras with source, status and measured capability; three onboarding paths.

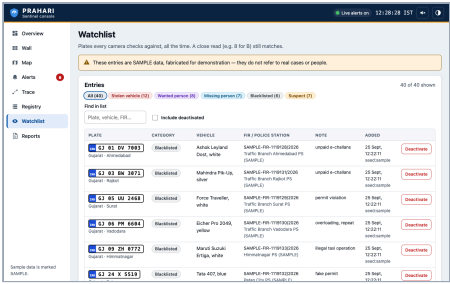


Figure 16.5 — Watchlist: categories, case details, CSV import; ships with a clearly labelled SAMPLE list.

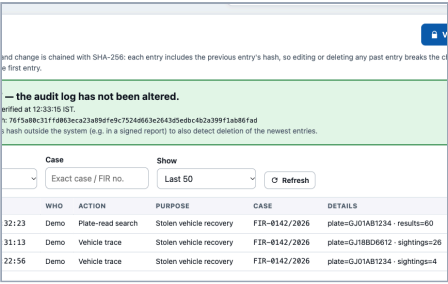


Figure 16.6 — Reports: plates CSV export, plate-read search and audit-chain verification.

16.4 Automated tests

148 tests, all passing (python -m pytest tests/ -q, 7.7 s, 25 September 2026) measured.

Test file	Tests	What is verified
test_node_catalogue.py	32	Tolerant catalogue parsing: wrapped lists, id-keyed objects, alias fields, nested/GeoJSON coordinates, resolutions, fractional fps, live words, URL discovery, duplicates
test_node_capture.py	11	Backoff ladder (capped, jittered, never tight); capture-time offset converges under GOP replay; discontinuity on backward/jump PTS but not on gaps; fps from PTS; PTS rate limiter on uneven sources; credential redaction; manager reconciliation, allow-list and cap
test_plate_grammar.py	20	Formats, State and RTO codes, constrained decoding, confusion distance, watchlist ranking and thresholds
test_anpr_pipeline.py	8	Event contract; synthetic accuracy floor; evidence crop size cap and disk-floor skip; IND-prefix and seam-duplicate repair; multi-frame pass de-duplicates to one event
test_anpr_twoline.py · test_anpr_dedup.py · test_anpr_ccap.py	5 · 9 · 5	Two-line split; per-pass clustering, epochs, best/first modes; capability verdicts and reasons
test_registry_alerts.py	18	Misreads alert on listed plates; different vehicle goes to review, unrelated plate creates nothing; idempotent ingest and alert de-duplication; batch ingest; acknowledgement; SSE replay and live push; purpose-bound watchlist search
test_registry_trace.py	9	Ordering and implausible-hop flagging; far-apart simultaneous sightings; matching via candidates; near-miss fallback never silent; grammar help; fuzzy search; plates CSV
test_registry_cameras.py	18	Catalogue sync over HTTP and inline; bulk CSV/JSON; manual entry, filters and GeoJSON; health merge; gap report
test_registry_audit.py	13	Purpose and case ID required on sensitive endpoints; sensitive queries and writes audited; append-only trigger; tampering detected (edited entry, deleted entry, single entry re-hashed); secrets redacted

Results on the ~50 government-provided feeds are reported separately in the Government-Feed Demonstration video and its plates output report, which are produced by the same code.

16.5 Limitations

These are the known gaps between the working system and a statewide production service. Each is scheduled in §15.2.

Limitation	Consequence today
Plate accuracy measured only on synthetic plates	Real CCTV (night, glare, blur, dirt, hand-painted fonts, angle) will be harder; no operational accuracy figure is claimed yet
Two-wheeler plates weak	Detector finds 13–18% of synthetic two-wheeler plates; two-line plates need ≥ 48 px
Small plates	Plates under ~24 px tall are not readable; such cameras are detected and reported, not fixed
No appearance re-identification	A trace breaks where the plate is unreadable; the gap is visible, not bridged
Operator authentication not enforced	Operator identity is self-asserted; suitable for a controlled demonstration, not open deployment
Single-process, SQLite reference build	Laptop-scale by design; production services, bus and PostgreSQL are designed, not built
Node link not yet production-grade	No mTLS gateway, disk store-and-forward or bandwidth ladder; events reach central ingest in-process
Evidence hashing and signed manifests not built	Audit log is tamper-evident; evidence crops are not yet individually hashed
Government-database adapters not built	Watchlists are loaded by CSV/API

Limitation	Consequence today
FRS and non-ANPR analytics not built	Non-traffic cameras get health monitoring and the video wall only
Wall shows snapshots	~2 s refresh of the latest frame; live-motion view is designed
Matcher and trace simplifications	Matching benchmarked at 200 watchlist entries; confusion costs hand-set; plausibility uses straight-line distance
Production hardware not benchmarked	Edge node classes in §12 are sized from M1 measurements with a ×2 safety factor

A Appendix A — API endpoint reference

Generated from the application's OpenAPI schema (`create_app().openapi()`, version 0.1.0) plus three live endpoints added by the runtime (`prahari/runtime.py`). Interactive documentation is served at `/docs`. Times accept unix seconds or ISO 8601 (naive = IST); responses carry unix `ts` and IST `*_iso`. Endpoints marked **PB** are purpose-bound: they require `purpose` and `case_id` and are audited. `X-Actor` carries the operator identity.

Method	Path	Tag	Summary
GET	<code>/api/health</code>	system	Service health and database stats
GET	<code>/api/cameras</code>	cameras	List / filter cameras (department, status, codec, source, live, bbox, q)
POST	<code>/api/cameras</code>	cameras	Onboarding path 3: add or update one camera manually
GET	<code>/api/cameras.geojson</code>	cameras	Cameras as GeoJSON (GIS layer)
POST	<code>/api/cameras/sync</code>	cameras	Onboarding path 1: sync from a catalogue URL or inline payload
POST	<code>/api/cameras/import</code>	cameras	Onboarding path 2: bulk import (CSV or JSON body)
GET	<code>/api/cameras/{camera_id}</code>	cameras	One camera
DELETE	<code>/api/cameras/{camera_id}</code>	cameras	Remove a camera (its plate events are kept)
POST	<code>/api/cameras/{camera_id}/health</code>	cameras	Heartbeat / measured capability from a node
GET	<code>/api/cameras/{camera_id}/snapshot.jpg</code>	live	Latest frame from a camera (JPEG, with plate boxes)
GET	<code>/api/watchlist</code>	watchlist	List watchlist entries
POST	<code>/api/watchlist</code>	watchlist	Add or update a watchlist entry
POST	<code>/api/watchlist/import</code>	watchlist	Bulk import (CSV body)
GET	<code>/api/watchlist/search PB</code>	watchlist	Rank watchlist entries against an observed plate
GET	<code>/api/watchlist/{plate}</code>	watchlist	One watchlist entry
DELETE	<code>/api/watchlist/{plate}</code>	watchlist	Deactivate an entry (soft delete)
POST	<code>/api/events/plate</code>	events	Ingest one PlateEvent (idempotent; screened; may raise alert/review)
POST	<code>/api/events/batch</code>	events	Ingest a batch of PlateEvents (store-and-forward flushes)
GET	<code>/api/events PB with plate</code>	events	Search plate events (fuzzy); latest events without plate
GET	<code>/api/alerts</code>	alerts	List alerts and review items
POST	<code>/api/alerts/{alert_id}/ack</code>	alerts	Acknowledge an alert, with note
GET	<code>/api/alerts/stream</code>	alerts	Live alerts (Server-Sent Events)
GET	<code>/api/trace/{plate} PB</code>	trace	Reconstruct a vehicle's route across cameras (JSON or GeoJSON)
GET	<code>/api/reports/gap</code>	reports	Coverage-gap analysis (JSON or Markdown)
GET	<code>/api/reports/plates.csv</code>	reports	Every detected plate with camera, location and timestamp (CSV; audited)
GET	<code>/api/audit</code>	audit	Recent audit entries
GET	<code>/api/audit/verify</code>	audit	Recompute and verify the hash chain
GET	<code>/api/runtime</code>	live	Live status: cameras, plate reader, throughput, disk
GET	<code>/api/evidence/{camera_id}/{filename}</code>	live	An evidence crop saved with a plate event

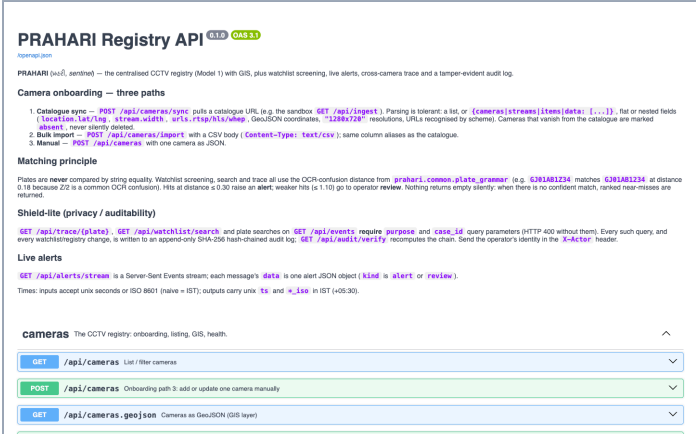


Figure A.1 — Interactive API documentation served by the running platform at /docs.

Request schemas: CameraIn, SyncIn, HealthIn, WatchlistIn, PlateEventIn, AckIn (prahari/api/models.py). Errors: 400 for validation and missing purpose binding (purpose_binding_required), 404 for unknown ids, 502 for an unreachable catalogue, 507 when the disk is below its safety floor.

B Appendix B — Data contracts and database schema

B.1 Data contracts (prahari/common/contracts.py)

CameraInfo — from a catalogue or adapter

Field	Type / meaning
id	str, stable camera id
name, department	str
lat, lon	float null
codec	"h264" "h265" ""
width, height	int null
fps	declared; never used for timing
live	bool
rtsp_url, hls_url, whep_url	str
raw	catalogue entry, verbatim

FrameSample — in memory only, never stored

Field	Type / meaning
camera_id	str
epoch	int; +1 on every reconnect or discontinuity
pts_s	presentation time, s
ts	estimated capture time, unix s (not arrival)
image	BGR uint8 array
width, height	int

PlateEvent — the unit that crosses the WAN

Field	Type / meaning
event_id	uuid4 hex; idempotency key
camera_id, ts, pts_s	as above
raw_text	OCR output, A–Z0–9
plate	best grammar-corrected plate or null
ocr_conf, det_conf	0..1
bbox	[x, y, w, h]
valid	raw text already a valid plate
candidates	[[plate, distance, score, edits]]
crop_path	evidence crop or null

Example PlateEvent: {"event_id": "4f1c0a8e...", "camera_id": "DEMO-04", "ts": 1790000000.0, "pts_s": 12.48, "raw_text": "GJ01AB1Z34", "plate": null, "ocr_conf": 0.71, "det_conf": 0.94, "bbox": [412, 380, 168, 44], "valid": false, "candidates": [{"plate": "GJ01AB1234", "distance": 0.18, "score": 0.98, "edits": ["pos7 Z->2"]}], "crop_path": null}

B.2 Database tables (prahari/registry/db.py)

Table	Columns	Keys, indexes, constraints
cameras	id, name, department, lat, lon, codec, width, height, fps, live, rtsp_url, hls_url, whep_url, source, status, last_seen, capability (JSON), raw (JSON), catalogue_url, created_at, updated_at	PK id; source ∈ {catalogue, manual, bulk, api}; indexes on department, status
plate_events	event_id, camera_id, ts, pts_s, raw_text, plate, ocr_conf, det_conf, bbox, valid, candidates (JSON), crop_path, received_at	PK event_id; indexes on plate, raw_text, (camera_id, ts), ts
watchlist	plate, category, details (JSON), source, active, added_at, updated_at	PK plate; category ∈ {stolen_vehicle, wanted_person, missing_person, blacklisted, suspect}

Table	Columns	Keys, indexes, constraints
alerts	id, event_id, watchlist_plate, observed, distance, score, via, kind, status, category, camera_id, ts, hits, last_ts, last_event_id, created, acked_by, acked_at, note	kind ∈ {alert, review}; status ∈ {open, ack}; indexes on (status, kind) and de-duplication key
alert_events	alert_id, event_id	PK (alert_id, event_id); every event folded into an alert
audit_log	id, ts, actor, action, purpose, case_id, params (JSON), prev_hash, hash	Append-only: triggers abort UPDATE and DELETE; SHA-256 chain

Production mapping: cameras → PostgreSQL + PostGIS (geometry column, spatial index); plate_events → TimescaleDB hypertable partitioned by day with (plate, ts) and trigram indexes; crops → object store keys; audit_log → append-only table with externally anchored head hash.

C Appendix C — Requirement traceability matrix

Every requirement in the official Problem Statement, with where it is answered and its status. Status: **BUILT** implemented and tested; **PARTLY** part built, rest designed; **DESIGNED**; **ROADMAP**.

ID	Requirement	Section	Status
C.1 HLD content (Submission Requirements, document 2)			
H1	Overall architecture, diagrams, component interactions	§4	BUILT (reference) / production DESIGNED
H2	Integrating heterogeneous cameras, NVRs and VMS	§5	PARTLY — RTSP/HLS/catalogue built; ONVIF/VMS adapters designed
H3	Ingest, process, manage streams from dispersed sites	§6	PARTLY — node built; WAN link, queue, ladder designed
H4	Watchlist integration and continuous correlation → real-time alerts	§7	BUILT ; government adapters DESIGNED
H5	AI analytics: ANPR, FRS, object detection, person/vehicle tracking, others	§8	ANPR, vehicle trace BUILT ; FRS DESIGNED ; others ROADMAP
H6	Alert workflow: prioritisation, visualisation, user interaction	§9	BUILT ; escalation DESIGNED
H7	Scalability, interoperability, security, performance for ~80,000 cameras	§5, §10, §12, §13, §16	Sized and designed; measured on reference build
H8	Technical prerequisites, assumptions, information required from departments	§2.4, §14	BUILT (measured capability) + checklist
C.2 Solution design dimensions (Step 3)			
D1–D3	Overall architecture · Integration strategy · AI & video analytics	§4 · §5–§7 · §8	As H1, H2/H4, H5
D4–D5	Cybersecurity architecture · Deployment architecture	§10 · §11	PARTLY · DESIGNED
D6–D7	Infrastructure sizing · Cost–benefit analysis	§12	Model built and reproducible (bench/sizing_model.py)
D8–D10	Department-wise information requirements · Scalability strategy · Future roadmap	§14 · §13 · §15	Answered
—	Architecture principles: open, modular, vendor-neutral, standard APIs, adapters, no lock-in	§2.3, §4.1, §5.2, App. A	BUILT (ONNX, OpenAPI, RTSP/HLS)
C.3 Plan for scale (Step 6)			
S1	Central, regional and edge compute requirements	§11, §12	Sized
S2	GPU / accelerator requirements	§12, §8.4	Sized from measured M1 cost
S3	Network bandwidth and low-bandwidth strategies	§3.3, §6.4, §12	Sized; ladder DESIGNED
S4	Hot, warm, cold storage by retention	§10.8, §12	Sized
S5	Load balancing, horizontal scaling, monitoring, logging, health checks	§13.1–13.2	Health BUILT ; rest DESIGNED
S6	HA, backup, DR, cybersecurity controls	§13.3–13.4, §10	DESIGNED
S7	Implementation and operational costs	§12	Estimated (±30%)
R1	Statewide rollout plan	§15	Answered
C.4 Model 1 — Registry & GIS (mandatory)			
M1.1	Bulk import, manual entry and API-based onboarding	§5.4	BUILT
M1.2	Interactive GIS map with department, camera type, status and coverage layers	§16.3	PARTLY — status, department, coverage built; camera type via API filter

ID	Requirement	Section	Status
M1.3	Camera health and maintenance-status monitoring	\$6.6, \$13.2	PARTLY — health built; AMC/maintenance fields designed
M1.4	Gap-analysis reports: uncovered zones, ageing infrastructure	\$14.2	PARTLY — uncovered zones, offline/stale built; install age designed
M1.5	Role-based search, filtering, export, metadata audit trails	\$10.3–10.5	PARTLY — search, filters, export, audit built; roles designed
M1.D	Deliverables: registry portal with GIS; bulk and manual onboarding demo; sample camera dataset; registry API docs; sample gap report	\$16, App. A	BUILT
C.5 Model 2 elements adopted			
M2.1	Feed aggregation through RTSP, ONVIF or vendor APIs	\$5.1	PARTLY — RTSP/HLS built
M2.2	ANPR-based metadata; event tagging; camera-wise indexing; searchable vehicle movements	\$7, \$8	BUILT
M2.3	Configurable video walls and multi-camera grid	\$6.6	BUILT (snapshot wall); live view DESIGNED
M2.4	Alerts for tagged events and vehicles of interest	\$9	BUILT
M2.D	Unified viewer on ≥ 2 systems; ANPR demo; searchable metadata dashboard; note that departmental systems remain unaffected	\$5.5, \$16	BUILT (Sentinel catalogue + own RTSP feed)
C.6 Model 3 elements adopted			
M3.1	Adapter/plugin architecture; extensible connector framework	\$5.2	PARTLY — contract and catalogue adapter built
M3.2	Metadata exchange bus for camera and event information	\$4.3	PARTLY — contracts and ingest API built; bus designed
M3.3	Cross-system event correlation; unified workflow and alert dashboard	\$7, \$9	BUILT
M3.D	Federating ≥ 2 systems; correlation dashboard; adapter documentation; federated analytics report	\$5.2, \$16	PARTLY — dashboard, reports, documentation built
C.7 Live technical evaluation (Step 4)			
T1	Onboard ~50 heterogeneous cameras onto one platform; centralised monitoring	\$5.3, \$6.6	BUILT (catalogue-driven)
T2	AI-powered video analytics on the grid	\$8	BUILT (ANPR, capability profiling)
T3	Given a plate: identify, trace, present movement; complete route with timestamped, location-wise history; GIS	\$8.6	BUILT
T4	Working watchlist database; continuous cross-referencing; automated real-time alerts	\$7, \$9	BUILT
T5	Output report of detected vehicles/plates with timestamps	App. A (plates.csv)	BUILT
C.8 Sandbox pre-submission checklist			
G1–G4	RTSP over TCP · no timing from declared fps or arrival · gaps do not stall · reconnect with backoff, tested by restarting a feed	\$5.6	BUILT , tested
G5–G8	Decoder warnings logged, not fatal · cameras and properties from /api/ingest · mixed H.264/H.265 and resolutions · sane across loop discontinuity	\$5.6	BUILT , tested
C.9 Government databases and watchlist entities			
W1	Watchlist entities: stolen vehicles, wanted persons, missing persons, blacklisted vehicles, suspects	\$7.1	BUILT (plate-keyed)
W2	Integration with VAHAN, SARTHI, eGujCop, AFIS, NAFIS	\$7.2	DESIGNED ; CSV/API import built
W3	Private cameras “wherever feasible and permitted”	\$10.9	DESIGNED
C.10 Bonus consideration			
B1	Innovative hybrid architecture with operational value	\$3, \$4	BUILT (reference) + sized
B2	Advanced cross-camera vehicle tracking / multi-camera correlation	\$8.6	BUILT ; Re-ID DESIGNED
B3	Additional reliable analytics beyond ANPR	\$8.5, \$8.8	Capability profiling BUILT ; others ROADMAP
B4	Edge processing, bandwidth optimisation, low-connectivity operation	\$6	Edge BUILT ; ladder/queue DESIGNED
B5	Cybersecurity, privacy, auditability, RBAC	\$10	Audit, purpose binding BUILT ; RBAC DESIGNED
B6	Operational dashboards, automated alerts, health monitoring, integration-ready APIs	\$9, \$13.2, App. A	BUILT

D Appendix D — Glossary and references

D.1 Glossary

Term	Meaning	Term	Meaning
ANPR	Automatic number-plate recognition	HSRP	High Security Registration Plate (with the blue “IND” strip)
CCAP	Camera capability profiling: measuring from a camera’s own frames whether it can support an analytic (here, ANPR)	mTLS	Mutual TLS: both client and server authenticate with certificates
Confusion distance	Edit distance in which substitutions that OCR commonly makes (O/0, l/1, B/8 ...) are cheap; PRAHARI’s replacement for string equality	Netram	District-level police command and control centres in Gujarat (34)
Constrained decoding	Correcting an OCR read by searching only among strings that are valid Indian plates	NVR / DVR / VMS	Network / digital video recorder; video management system
Epoch	A continuous stretch of one camera’s stream; a new epoch starts on every reconnect or timestamp discontinuity	PTS	Presentation timestamp carried in the video stream
GOP / IDR	Group of pictures / instantaneous decoder refresh (keyframe) in H.264/H.265	Re-ID	Re-identification of the same vehicle across cameras by appearance
HLS, RTSP, ONVIF, WHIP	HTTP Live Streaming; Real Time Streaming Protocol; Open Network Video Interface Forum device standard; WebRTC-HTTP egress protocol	VISWAS	Video Integration and State-Wide Advanced Security, Gujarat’s police CCTV programme

D.2 References

1. Gujarat Police Sentinel Hackathon 2026 — Problem Statement and Solution Flow; Resources (sandbox notes); FAQs. Official portal, accessed September 2026.
2. VISWAS Phase I: DeshGujarat, 2 July 2022, “7000 CCTV cameras installed in Gujarat under phase-I of VISWAS”. Phase II: DeshGujarat, 15 December 2023, “10,500 CCTV cameras being installed across Gujarat under VISWAS project”.
3. UK Home Office, National ANPR Service — Data Protection Impact Assessment (GOV.UK); Leonardo UK, “Driven by innovation: the UK National ANPR Service”; NPCC ANPR Strategy 2020.
4. Axon, “Fusus unified interface” (product literature). Seoul Metropolitan Government, “Seoul CCTV Integrated Control Center”; Korea Personal Information Protection Act, Art. 25.
5. The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023), Government of India.
6. *Justice K.S. Puttaswamy (Retd.) v. Union of India*, (2017) 10 SCC 1.
7. Open-source components: FFmpeg/PyAV, ONNX Runtime, open-image-models, fast-plate-ocr, FastAPI, Leaflet, OpenStreetMap. Price and hardware anchors: §12.11.